Name

Mathematics Advanced Year 12

Integral Calculus

Book 1

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Syllabus Content

MAV-12-05 solves problems involving indefinite and definite integrals

Indefinite integrals

- Use the notation $\int f(x)dx$ for the general primitive of $f(x)$, called the indefinite integral of $f(x)$, so that $\int f(x)dx = F(x) + C$, for some constant C, where $F(x)$ is any primitive of $f(x)$
- Recognise integration as the process of finding the indefinite integral of a function
- Use the formula $\int x^n dx = \frac{1}{n+1}x^{n+1} + C$ for real $n \neq -1$
- Use the identities $\int \big(f(x) + g(x)\big)dx = \int f(x)dx + \int g(x)dx$ and $\int kf(x)dx = k\int f(x)dx$ for primitives
- Prove by differentiation, and apply $\int u^n \frac{du}{dx}dx = \frac{1}{n+1}u^{n+1} + C$, where u is a function of x, or $\int f'(x)[f(x)]^n dx = \frac{1}{n+1}[f(x)]^{n+1} + C$, for real $n \neq -1$

Integration with exponential functions

- Establish and use the formula $\int e^x dx = e^x + C$
- Establish and use the formula $\int e^{ax+b}dx = \frac{1}{a}e^{ax+b} + C$, where a and b are constants and $a \neq 0$
- Establish and use the formula $\int a^x dx = \frac{a^x}{\ln a} + C$, where a is a constant and $a > 0$
- Establish and use $\int e^u \frac{du}{dx}dx = e^u + C$, where u is a function of x, or $\int f'(x)e^{f(x)}dx = e^{f(x)} + C$
- Find primitives of functions involving exponential functions

Integration with logarithmic functions

- Derive and use the formula $\int \frac{1}{x}dx = \ln|x| + C$ where $x \neq 0$
- Establish and use the formula $\int \frac{1}{ax+b}dx = \frac{1}{a}\ln|ax+b| + C$, where a and b are constants and $a \neq 0$
- Establish and use $\int \frac{u'}{u}dx = \ln|u| + C$, where u is a function of x, or $\int \frac{f'(x)}{f(x)}dx = \ln|f(x)| + C$, on a domain where $f(x) \neq 0$

Integration with trigonometric functions

- Establish and use the formulas $\int \sin x\,dx = -\cos x + C$, $\int \cos x\,dx = \sin x + C$ and $\int \sec^2 x\,dx = \tan x + C$
- Establish and use indefinite integrals of the form $\int f(ax+b)dx$, where a and b are constants and $a \neq 0$, and $f(x) = \sin x$, $f(x) = \cos x$ and $f(x) = \sec^2 x$
- Determine indefinite integrals of the form $\int f'(x)\sin f(x)\,dx$, $\int f'(x)\cos f(x)\,dx$ and $\int f'(x)\sec^2 f(x)\,dx$

Integrating Power Functions

ⓘ Indefinite Integral

Because of the close relationship between primitives and the definite integral, we call the **general primitive** the **indefinite integral**. We write it using an integral sign without upper or lower limits.
For example:

$$\int x^2\,dx = \frac{x^3}{3} + C$$

A **definite integral is a number** that represents the area under a curve.
An **indefinite integral is an expression** that represents the general primitive function.

We call the process of finding the primitive **integration** and use the term '**integral**' to refer to both the definite and indefinite integrals.
The expression being integrated is called the **integrand**.

ⓘ Integrals of Power Functions

$$\int x^n\,dx = \frac{x^{n+1}}{n+1} + C$$

$$\int (ax+b)^n\,dx = \frac{(ax+b)^{n+1}}{a(n+1)} + C$$

⚠ Remember to expand any brackets and split any fractions.
⚠ We can factor any constant outside the integral: $\int kf(x)\,dx = k\int f(x)dx$

Example Find the indefinite integral of power functions (basic)

$$\int 9\,dx = 9x + C$$

$$\int 12x^3\,dx = \frac{12x^4}{4} + C$$
$$= 3x^4 + C$$

Guided Practice

a

$$\int 8x^5\,dx$$

$\frac{4x^6}{3} + C$

b

$$15\int x^2\,dx$$

$5x^3 + C$

Example Find the indefinite integral of power functions (linear chain rule)

$$\int (3x+1)^5\,dx = \frac{(3x+1)^6}{3(6)} + C$$
$$= \frac{(3x+1)^6}{18} + C$$

$$\int (5-2x)^2\,dx = \frac{(5-2x)^3}{-2(3)} + C$$
$$= \frac{(5-2x)^3}{-6} + C$$

Guided Practice

a

$$\int (2x+7)^4\,dx$$

$\frac{(2x+7)^5}{10} + C$

b

$$\int (4-3x)^6\,dx$$

$-\frac{(4-3x)^7}{21} + C$

Example Find the indefinite integral of power functions (negative powers)

$$\int \frac{12}{x^3}\,dx = \int 12x^{-3}\,dx$$
$$= \frac{12x^{-2}}{-2} + C$$
$$= -\frac{6}{x^2} + C$$

$$\int \frac{dx}{(3x+4)^2} = \int (3x+4)^{-2}dx$$
$$= \frac{(3x+4)^{-1}}{3(-1)} + C$$
$$= -\frac{1}{3(3x+4)} + C$$

Guided Practice

a

$$\int \frac{20}{x^5}\,dx$$

$-\frac{5}{x^4} + C$

b

$$\int \frac{dx}{(1-4x)^2}$$

$\frac{1}{4(1-4x)} + C$

Example Find the indefinite integral of power functions (expand/split first)

$$\int (x^3-1)^2\,dx = \int x^6 - 2x^3 + 1\,dx$$
$$= \frac{x^7}{7} - \frac{2x^4}{4} + x + C$$
$$= \frac{x^7}{7} - \frac{x^4}{2} + x + C$$

$$\int \frac{6x^4 - 8x^2 + 3}{2x^2}\,dx = \int 3x^2 - 4 + \frac{3}{2}x^{-2}\,dx$$
$$= x^3 - 4x + \frac{3x^{-1}}{2(-1)} + C$$
$$= x^3 - 4x - \frac{3}{2x} + C$$

Guided Practice

a

$$\int (x^2+3)^2\,dx$$

$\frac{x^5}{5} + 2x^3 + 9x + C$

b

$$\int \frac{10x^3 + 4x}{2x}\,dx$$

$\frac{5x^3}{3} + 2x + C$

Example Find the indefinite integral of power functions (fractional indices)

$$\int \sqrt{x}\,dx = \int x^{\frac{1}{2}}\,dx$$
$$= \frac{x^{\frac{3}{2}}}{\left(\frac{3}{2}\right)} + C$$
$$= \frac{2\sqrt{x^3}}{3} + C$$

$$\int \frac{1}{\sqrt{9-2x}}\,dx = \int (9-2x)^{-\frac{1}{2}}\,dx$$
$$= \frac{(9-2x)^{\frac{1}{2}}}{-2\left(\frac{1}{2}\right)} + C$$
$$= -\sqrt{9-2x} + C$$

Guided Practice

a

$$\int \sqrt[3]{x}\,dx$$

$\frac{3\sqrt[3]{x^4}}{4} + C$

b

$$\int \sqrt{7-2x}\,dx$$

$-\frac{\sqrt{(7-2x)^3}}{3} + C$

1 Find these indefinite integrals.

a $\int 4\,dx$ $4x + C$

b $\int 1\,dx$ $x + C$

c $\int 0\,dx$ C

d $\int (-2)\,dx$ $-2x + C$

e $\int x\,dx$ $\frac{x^2}{2} + C$

f $\int x^2\,dx$ $\frac{x^3}{3} + C$

g $\int x^3\,dx$ $\frac{x^4}{4} + C$

h $\int x^7\,dx$ $\frac{x^8}{8} + C$

2 Find these indefinite integrals.

a $\int (x + x^2)\,dx$ $\frac{x^2}{2} + \frac{x^3}{3} + C$

b $\int (x^4 - x^3)\,dx$ $\frac{x^5}{5} - \frac{x^4}{4} + C$

c $\int (x^7 + x^{10})\,dx$ $\frac{x^8}{8} + \frac{x^{11}}{11} + C$

d $\int (2x + 5x^4)\,dx$ $x^2 + x^5 + C$

e $\int (9x^8 - 11)\,dx$ $x^9 - 11x + C$

f $\int (7x^{13} + 3x^8)\,dx$ $\frac{x^{14}}{2} + \frac{x^9}{3} + C$

g $\int (4 - 3x)\,dx$ $4x - \frac{3x^2}{2} + C$

h $\int (1 - x^2 + x^4)\,dx$ $x - \frac{x^3}{3} + \frac{x^5}{5} + C$

i $\int (3x^2 - 8x^3 + 7x^4)\,dx$ $x^3 - 2x^4 + \frac{7x^5}{5} + C$

3 Find the indefinite integrals.

a $\int x^{-2}\,dx$ $\frac{x^{-1}}{-1} = -x^{-1} + C$

b $\int x^{-3}\,dx$ $\frac{x^{-2}}{-2} = -\left(\frac{1}{2}\right)x^{-2} + C$

c $\int x^{-8}\,dx$ $\frac{x^{-7}}{-7} = -\left(\frac{1}{7}\right)x^{-7} + C$

d $\int 3x^{-4}\,dx$ $3 \cdot \frac{x^{-3}}{-3} = -x^{-3} + C$

e $\int 9x^{-10}\,dx$ $9 \cdot \frac{x^{-9}}{-9} = -x^{-9} + C$

f $\int 10x^{-6}\,dx$ $10 \cdot \frac{x^{-5}}{-5} = -2x^{-5} + C$

4 Find the indefinite integrals.

a $\int x^{\frac{1}{2}}\,dx$

$\frac{x^{\frac{3}{2}}}{\frac{3}{2}} = \frac{2}{3}x^{\frac{3}{2}} + C$

b $\int x^{\frac{1}{3}}\,dx$

$\frac{x^{\frac{4}{3}}}{\frac{4}{3}} = \frac{3}{4}x^{\frac{4}{3}} + C$

c $\int x^{\frac{1}{4}}\,dx$

$\frac{x^{\frac{5}{4}}}{\frac{5}{4}} = \frac{4}{5}x^{\frac{5}{4}} + C$

d $\int x^{\frac{2}{3}}\,dx$

$\frac{x^{\frac{5}{3}}}{\frac{5}{3}} = \frac{3}{5}x^{\frac{5}{3}} + C$

e $\int x^{-\frac{1}{2}}\,dx$

$\frac{x^{\frac{1}{2}}}{\frac{1}{2}} = 2x^{\frac{1}{2}} + C$

f $\int 4x^{\frac{1}{2}}\,dx$

$4 \cdot \frac{x^{\frac{3}{2}}}{\frac{3}{2}} = \frac{8}{3}x^{\frac{3}{2}} + C$

5 Find the indefinite integrals.

a $\int x(x + 2)\,dx$

$\frac{x^3}{3} + x^2 + C$

b $\int x(4 - x^2)\,dx$

$2x^2 - \frac{x^4}{4} + C$

c $\int x^2(5 - 3x)\,dx$

$\frac{5x^3}{3} - \frac{3x^4}{4} + C$

d $\int x^3(x - 5)\,dx$

$\frac{x^5}{5} - \frac{5x^4}{4} + C$

e $\int (x - 3)^2\,dx$

$\frac{x^3}{3} - 3x^2 + 9x + C$

f $\int (2x + 1)^2\,dx$

$\frac{4x^3}{3} + 2x^2 + x + C$

g $\int (1 - x^2)^2\,dx$

$x - \frac{2x^3}{3} + \frac{x^5}{5} + C$

h $\int (2 - 3x)(2 + 3x)\,dx$

$4x - 3x^3 + C$

i $\int (x^2 - 3)(1 - 2x)\,dx$

$\frac{x^3}{3} - \frac{x^4}{2} - 3x + 3x^2 + C$

6 Find the indefinite integrals.

a $\int \frac{x^2 + 2x}{x}\,dx$

$\frac{x^2}{2} + 2x + C$

b $\int \frac{x^7 + x^8}{x^6}\,dx$

$\frac{x^2}{2} + \frac{x^3}{3} + C$

c $\int \frac{2x^3 - x^4}{4x}\,dx$

$\frac{x^3}{6} - \frac{x^4}{16} + C$

7 Find the indefinite integrals.

a $\int x^{-2}\,dx$

$-\frac{1}{x}+C$

b $\int x^{-3}\,dx$

$-\frac{1}{2x^2}+C$

c $\int x^{-5}\,dx$

$-\frac{1}{4x^4}+C$

d $\int x^{-10}\,dx$

$-\frac{1}{9x^9}+C$

e $\int 3x^{-4}\,dx$

$-\frac{1}{x^3}+C$

f $\int 5x^{-6}\,dx$

$-\frac{1}{x^5}+C$

g $\int 7x^{-8}\,dx$

$-\frac{1}{x^7}+C$

h $\int \frac{1}{3}x^{-2}\,dx$

$-\frac{1}{3x}+C$

i $\int \frac{x^{-5}}{7}dx$

$-\frac{1}{28x^4}+C$

j $-\frac{1}{5}\int x^{-3}\,dx$

$\frac{1}{10x^2}+C$

k $\int (x^{-2}-x^{-5})\,dx$

$\frac{1}{4x^4}-\frac{1}{x}+C$

l $\int (x^{-3}+x^{-4})\,dx$

$-\frac{1}{2x^2}-\frac{1}{3x^3}+C$

8 Find the indefinite integrals.

a $\int \sqrt{x}\,dx$

$\left(\frac{2}{3}\right)x^{\frac{3}{2}}+C$

b $\int \sqrt[3]{x}\,dx$

$\left(\frac{3}{4}\right)x^{\frac{4}{3}}+C$

c $\int \frac{1}{\sqrt{x}}\,dx$

$2\sqrt{x}+C$

d $\int \sqrt[3]{x^2}\,dx$

$\left(\frac{3}{5}\right)x^{\frac{5}{3}}+C$

9 Integrate:

a $\int (x+1)^5\,dx$

$\left(\frac{1}{6}\right)(x+1)^6+C$

b $\int (x+2)^3\,dx$

$\left(\frac{1}{4}\right)(x+2)^4+C$

c $\int (4-x)^4\,dx$

$-\left(\frac{1}{5}\right)(4-x)^5+C$

d $\int (3-x)^2\,dx$

$-\left(\frac{1}{3}\right)(3-x)^3+C$

e $\int (3x+1)^4\,dx$

$\left(\frac{1}{15}\right)(3x+1)^5+C$

f $\int (4x-3)^7\,dx$

$\left(\frac{1}{32}\right)(4x-3)^8+C$

g $\int (5-2x)^6\,dx$

$-\left(\frac{1}{14}\right)(5-2x)^7+C$

h $\int (1-5x)^7\,dx$

$-\left(\frac{1}{40}\right)(1-5x)^8+C$

i $\int (2x+9)^{11}\,dx$

$\left(\frac{1}{24}\right)(2x+9)^{12}+C$

j $\int 3(2x-1)^{10}\,dx$

$\left(\frac{3}{22}\right)(2x-1)^{11}+C$

k $\int 4(5x-4)^6\,dx$

$\left(\frac{4}{35}\right)(5x-4)^7+C$

l $\int 7(3-2x)^3\,dx$

$-\left(\frac{7}{8}\right)(3-2x)^4+C$

m $\int \left(\frac{x}{3} - 7\right)^4 dx$ $\left(\frac{3}{5}\right)\left(\frac{x}{3} - 7\right)^5 + C$	**n** $\int \left(\frac{x}{4} - 7\right)^6 dx$ $\left(\frac{4}{7}\right)\left(\frac{x}{4} - 7\right)^7 + C$	**o** $\int \left(1 - \frac{x}{5}\right)^3 dx$ $-\left(\frac{5}{4}\right)\left(1 - \frac{x}{5}\right)^4 + C$
p $\int \frac{1}{(x+1)^3} dx$ $-\frac{1}{2(x+1)^2} + C$	**q** $\int \frac{1}{x - 5^4} dx$ $-\frac{1}{3(x-5)^3} + C$	**r** $\int \frac{1}{(3x-4)^2} dx$ $-\frac{1}{3(3x-4)} + C$
s $\int \frac{1}{(2-x^5)} dx$ $\frac{1}{4(2-x)^4} + C$	**t** $\int \frac{3}{(x-7)^6} dx$ $-\frac{3}{5(x-7)^5} + C$	**u** $\int \frac{8}{(4x+1)^5} dx$ $-\frac{1}{2(4x+1)^4} + C$
v $\int \frac{2}{(3-5x)^4} dx$ $\frac{2}{15(3-5x)^3} + C$	**w** $\int \frac{4}{5(1-4x)^2} dx$ $= \frac{1}{5-20x} + C$	**x** $\int \frac{7}{8(3x+2)^5} dx$ $-\frac{7}{96(3x+2)^4} + C$
y $\int \sqrt{x}(3\sqrt{x} - x)\, dx$ $\frac{3x^2}{2} - \left(\frac{2}{5}\right)x^{\frac{5}{2}} + C$	**z** $\int (\sqrt{x} - 2)(\sqrt{x} + 2)\, dx$ $\frac{x^2}{2} - 4x + C$	**aa** $\int (2\sqrt{x} - 1)^2\, dx$ $2x^2 - \left(\frac{8}{3}\right)x^{\frac{3}{2}} + x + C$

10 Explain why the indefinite integral $\int \frac{1}{x}\, dx$ cannot be found using the standard form $\int x^n\, dx = \frac{x^{n+1}}{n+1} + C$.

$\int x^{-1}\, dx$ using the rule gives $\frac{x^0}{0} + C$

which is meaningless (division by zero). The rule $\int x^n\, dx = \frac{x^{n+1}}{n+1} + C$ requires $n \neq -1$

11 Evaluate these definite integrals.

a $\int_0^4 \sqrt{x}\,dx$

$\frac{16}{3}$

b $\int_1^9 x\sqrt{x}\,dx$

$\frac{484}{5}$

c $\int_1^9 \frac{1}{\sqrt{x}}dx$

4

d $\int_2^4 (2-\sqrt{x})(2+\sqrt{x})\,dx$

2

e $\int_0^1 \sqrt{x}(\sqrt{x}-4)\,dx$

$-\frac{13}{6}$

f $\int_4^9 (\sqrt{x}-1)^2\,dx$

$\frac{73}{6}$

12 Find each indefinite integral.

a $\int \sqrt{2x-1}\,dx$

$\left(\frac{1}{3}\right)(2x-1)^{\frac{3}{2}}+C$

b $\int \sqrt{7-4x}\,dx$

$-\left(\frac{1}{6}\right)(7-4x)^{\frac{3}{2}}+C$

c $\int \sqrt[3]{4x-1}\,dx$

$\left(\frac{3}{16}\right)(4x-1)^{\frac{4}{3}}+C$

d $\int \frac{1}{\sqrt{3x+5}}dx$

$\left(\frac{2}{3}\right)\sqrt{3x+5}+C$

13 Evaluate these definite integrals.

a $\int_0^2 (x+1)^4\,dx$

$\frac{242}{5}$

b $\int_2^3 (2x-5)^3\,dx$

0

c $\int_{-2}^2 (1-x)^5\,dx$

$\frac{364}{3}$

d $\int_0^5 \left(1-\frac{x}{5}\right)^4 dx$

1

e $\int_0^1 \sqrt{9-8x}\,dx$

$\frac{13}{6}$

f $\int_2^7 \frac{1}{\sqrt{x+2}}dx$

2

g $\int_{-2}^0 \sqrt[3]{x+1}\,dx$

0

h $\int_1^5 \sqrt{3x+1}\,dx$

$\frac{112}{9}$

i $\int_{-3}^0 \sqrt{1-5x}\,dx$

$\frac{42}{5}$

Properties of the Definite Integral

Review

Classify functions as odd, even, or neither

a	$f(x) = x^5$	**b**	$f(x) = x^2 + 1$	**c**	$f(x) = \sin x$
d	$f(x) = \cos x$	**e**	$f(x) = x^3 + x$	**f**	$f(x) = x^3 + 1$
g	$f(x) = x^4 - x^2$	**h**	$f(x) = \frac{1}{x}$	**i**	$f(x) = \sin 2x$

How do you know whether a polynomial is odd?

How do you know whether a polynomial is even?

The sum of two odd functions is ……

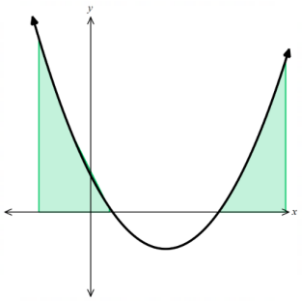
The sum of two even functions is ……

ⓘ The Sign of the Definite Integral

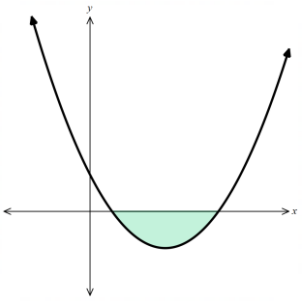
The definite integral returns a positive value when the function is above the x-axis and a negative value when the function is below the x-axis.

https://www.desmos.com/calculator/zvuixbav1g

Positive Area

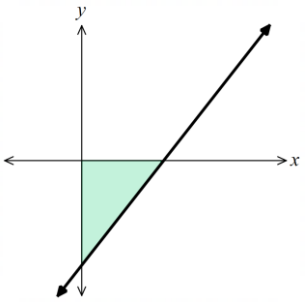


Negative Area

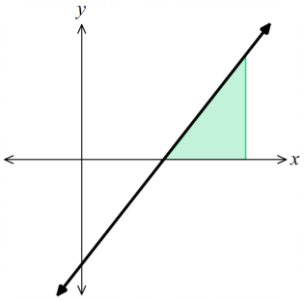


Example Interpret the sign of an integral

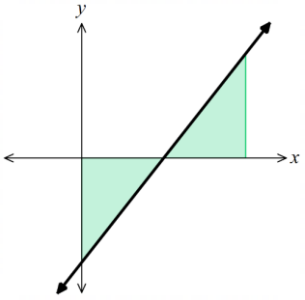
$$\int_0^3 (2x-6)\,dx = \Big[x^2-6x\Big]_0^3$$
$$= (9-18)-(0-0)$$
$$= -9$$



$$\int_3^6 (2x-6)\,dx = \Big[x^2-6x\Big]_3^6$$
$$= (36-36)-(9-18)$$
$$= 9$$



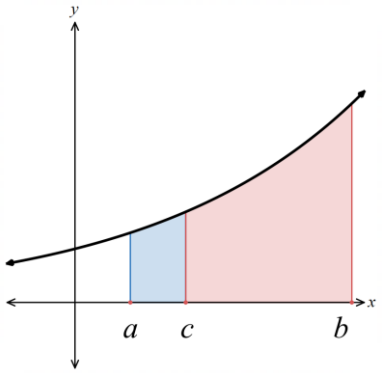
$$\int_0^6 (2x-6)\,dx = \Big[x^2-6x\Big]_0^6$$
$$= (36-36)-(0-0)$$
$$= 0$$



ⓘ Additive Property of Intervals

You can split a region at any intermediate point and add the subintervals.

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$$



ⓘ **Apply** additive property of integrals

a Given $\int_0^3 f(x)\,dx = 15$ and $\int_3^8 f(x)\,dx = 9$, find $\int_0^8 f(x)\,dx$.

b Given $\int_1^4 f(x)\,dx = 14$ and $\int_4^6 f(x)\,dx = 5$, find $\int_1^6 f(x)\,dx$.

c Given $\int_2^7 f(x)\,dx = 10$ and $\int_2^5 f(x)\,dx = 13$, find $\int_5^7 f(x)\,dx$.

d Given $\int_0^4 f(x)\,dx = -6$ and $\int_0^2 f(x)\,dx = 3$, find $\int_2^4 f(x)\,dx$.

ⓘ Reversal of Limits Property

Swapping the limits of integration negates the result.

$$\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$$

This is because $[F(b) - F(a)] = -[F(a) - F(b)]$

ⓘ **Apply** reversal of limits property of the definite integral

a Given $\int_2^7 f(x)\,dx = 11$,
find $\int_7^2 f(x)\,dx$.

b Given $\int_5^1 f(x)\,dx = -3$,
find $\int_1^5 f(x)\,dx$

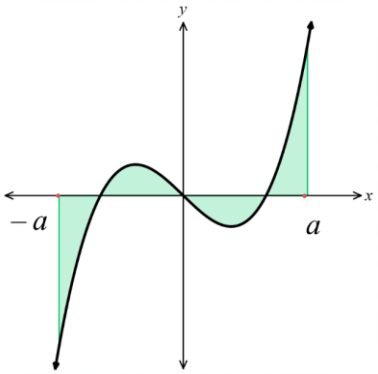
c Given $\int_0^6 f(x)\,dx = 20$ and
$\int_4^0 f(x)\,dx = -9$,
find $\int_4^6 f(x)\,dx$.

d Given $\int_3^7 f(x)\,dx = 5$ and
$\int_0^7 f(x)\,dx = 18$,
Find $\int_3^0 f(x)\,dx$

ⓘ The Definite Integral and Symmetries

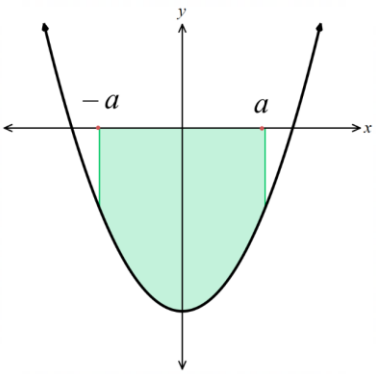
For an **odd function**, the area above the x-axis on one side is exactly cancelled by the area below on the other side.

$$\int_{-a}^{a} f(x)\,dx = 0$$



For an **even function**, the area from $-a$ to 0 is identical to the area from 0 to a.
You can evaluate one half and double it.

$$\int_{-a}^{a} f(x)\,dx = 2\int_{0}^{a} f(x)\,dx$$



ⓘ **Apply** symmetry properties of the definite integral

a Given $f(x)$ is even and $\int_0^3 f(x)\,dx = 8$, find $\int_{-3}^{3} f(x)\,dx$.

b Given $g(x)$ is odd and $\int_0^5 g(x)\,dx = 8$, find $\int_{-5}^{5} g(x)\,dx$.

c Given $f(x)$ is even and $\int_{-2}^{2} f(x)\,dx = 14$, find $\int_0^2 f(x)\,dx$.

d Given $g(x)$ is odd and $\int_{-2}^{5} g(x)\,dx = 8$, find $\int_2^5 g(x)\,dx$.

The Definite Integral and Symmetries

When the limits of an integral are from $-a$ to a, check whether the function is odd or even.

⚠ Also test whether you are integrating across an asymptote.

For an **odd function**:

$$\int_{-a}^{a} f(x)\,dx = 0$$

For an **even function**:

$$\int_{-a}^{a} f(x)\,dx = 2\int_{0}^{a} f(x)\,dx$$

Example Integrate using symmetry properties of the definite integral

$$\int_{-2}^{2} (x^3 - 4x)\,dx$$

This is an odd function.

$$\int_{-2}^{2} (x^3 - 4x)\,dx = 0$$

$$\int_{-2}^{2} (x^2 + 1)\,dx$$

This is an even function.

$$2\int_{0}^{2} (x^2 + 1)\,dx$$

$$= 2\left[\frac{x^3}{3} + x\right]_0^2$$

$$= 2\left(\left(\frac{8}{3} + 2\right) - (0 + 0)\right)$$

$$= \frac{28}{3}$$

Guided Practice

a

$$\int_{-2}^{2} (x^4 + x^2 + 3)\,dx$$

$\frac{452}{15}$

b

$$\int_{-2}^{2} \frac{7}{x}\,dx$$

undefined

c

$$\int_{-5}^{5} \left(\sin x + xe^{x^2}\right)\,dx$$

0

d

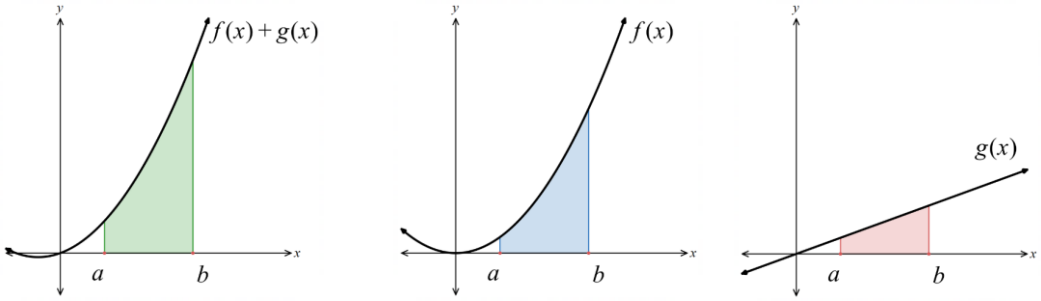
$$\int_{-\frac{\pi}{3}}^{\frac{\pi}{3}} \tan x\,dx$$

0

Integral of a Sum or Difference of Functions

You can break a complicated integrand into simpler pieces and integrate each separately. Useful when integrating a sum of an even and odd function and the limits are symmetric.

$$\int_a^b \big(f(x)+g(x)\big)\,dx = \int_a^b f(x)\,dx + \int_a^b g(x)\,dx$$



Example Integrate sums or differences of functions using reverse chain rule

$$\int_{-3}^{3}(x^4+x^3)\,dx = \int_{-3}^{3} x^4\,dx + \int_{-3}^{3} x^3\,dx$$

$$= 2\int_0^3 x^4\,dx + 0 \text{ because } x^4 \text{ is even and } x^3 \text{ is odd}$$

$$= 2\left[\frac{x^5}{5}\right]_0^3$$

$$= 2\left(\frac{243}{5} - 0\right)$$

$$= \frac{486}{5}$$

Guided Practice

a

$$\int_{-2}^{2} x^2 + x^5\,dx$$

$\frac{16}{3}$

b

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\sin 2x + \cos\frac{x}{3}\right) dx$$

3

1 Evaluate $\int_4^5 (2x - 3)dx$ and $\int_5^4 (2x - 3)dx$. What do you notice?

$$\int_4^5 (2x - 3)dx = [x^2 - 3x]_4^5 = (25 - 15) - (16 - 12) = 6$$

$$\int_5^4 (2x - 3)dx = [x^2 - 3x]_5^4 = (16 - 12) - (25 - 15) = -6$$

The values are 6 and −6. Reversing the limits changes the sign.

2 Show, by evaluating the definite integrals, that:

a $\int_0^1 6x^2 dx = 6\int_0^1 x^2 dx$

$$LHS: \int_0^1 6x^2 dx = [2x^3]_0^1 = 2$$

$$RHS: 6\int_0^1 x^2 dx = 6\left[\frac{x^3}{3}\right]_0^1 = 6\left(\frac{1}{3}\right) = 2$$

$$LHS = RHS = 2$$

b $\int_{-1}^2 (x^3 + x^2)dx = \int_{-1}^2 x^3 dx + \int_{-1}^2 x^2 dx$

$$LHS: \int_{-1}^2 (x^3 + x^2)dx = \left[\frac{x^4}{4} + \frac{x^3}{3}\right] - 1^2 = \left(4 + \frac{8}{3}\right) - \left(\frac{1}{4} - \frac{1}{3}\right) = \frac{27}{4}$$

$$RHS: \int_{-1}^2 x^3 dx + \int_{-1}^2 x^2 dx = \left[\frac{x^4}{4}\right] - 1^2 + \left[\frac{x^3}{3}\right]_{-1}^2 = \left(4 - \frac{1}{4}\right) + \left(\frac{8}{3} + \frac{1}{3}\right) = \frac{15}{4} + 3 = \frac{27}{4}$$

$$LHS = RHS = \frac{27}{4}$$

c $\int_0^3 (x^2 - 4x + 3)dx = \int_0^2 (x^2 - 4x + 3)dx + \int_2^3 (x^2 - 4x + 3)dx$

$$LHS: \int_0^3 (x^2 - 4x + 3)dx = \left[\frac{x^3}{3} - 2x^2 + 3x\right]_0^3 = (9 - 18 + 9) - 0 = 0$$

$$RHS: \int_0^2 (x^2 - 4x + 3)dx + \int_2^3 (x^2 - 4x + 3)dx$$

$$= \left[\frac{x^3}{3} - 2x^2 + 3x\right]_0^2 + \left[\frac{x^3}{3} - 2x^2 + 3x\right]_2^3$$

$$= \left(\frac{8}{3} - 8 + 6\right) + \left((9 - 18 + 9) - \left(\frac{8}{3} - 8 + 6\right)\right)$$

$$= 0$$

$$LHS = RHS = 0$$

3 Without evaluating the definite integrals, explain why:

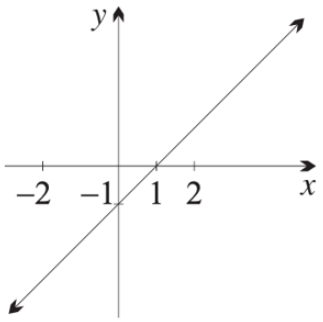
a $\int_2^2 (x^2 - 3x)dx = 0$

The upper and lower limits are equal, so the interval has zero width. Therefore $\int_2^2 (x^2 - 3x)dx = 0$.

b $\int_{-2}^2 x\,dx = 0$

$y = x$ is an odd function, so the areas above and below the x-axis on [−2, 2] cancel. Therefore $\int_{-2}^2 x\,dx = 0$

4 The diagram shows the line $y = x - 1$.



Without evaluating the definite integrals, explain why:

a $\int_0^1 (x - 1)\,dx$ is negative

On [0, 1], the line $y = x - 1$ lies below the x-axis.
The integral is negative.

b $\int_1^2 (x - 1)\,dx$ is positive

On [1, 2], the line $y = x - 1$ lies above the x-axis.
The integral is positive

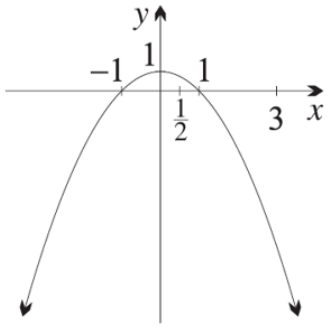
c $\int_0^2 (x - 1)\,dx$ is zero

On [0, 1] the line is below the x-axis, on [1, 2] it is above.
The two triangles have equal area (both base 1, height 1), so the signed areas cancel. The integral is zero.

d $\int_{-2}^2 (x - 1)\,dx$ is negative

On [−2, 1] the line is below the x-axis
On [1, 2] it is above
The negative area exceeds the positive area.
The integral is negative.

5 The diagram shows the parabola $y = 1 - x^2$.



Without evaluating the definite integrals, explain why:

a $\int_{-1}^1 (1 - x^2)dx$ is positive

On [−1, 1], the parabola is above the x-axis. The integral is positive.

b $\int_1^3 (1 - x^2)dx$ is negative

c

On [1, 3], the parabola is below the x-axis. The integral is negative.

d $\int_{-1}^0 (1 - x^2)dx = \int_0^1 (1 - x^2)dx$

$y = 1 - x^2$ is an even function,
so it is symmetrical about the y-axis.
The area from −1 to 0 equals the area from 0 to 1.

e $\int_0^{\frac{1}{2}} (1 - x^2)dx > \int_{\frac{1}{2}}^1 (1 - x^2)\,dx$

On [0, 1], the curve y = 1 − x² is positive and decreasing.
The curve is higher on [0, 1/2] than on [1/2, 1], so the area under the parabola from 0 to 1/2 is greater than the area from 1/2 to 1.

6

a If $\int_1^3 f(x)dx = 7$, then what is the value of $\int_3^1 f(x)dx$?

Reversing the limits changes the sign. −7

b If $\int_2^1 g(x)dx = -5$, then what is the value of $\int_1^2 g(x)dx$?

Reversing the limits changes the sign. 5

7 In each part, evaluate the definite integrals.
Then use the properties of the definite integral to explain the relationship.

a $\int_0^2 (3x^2 - 1)dx$

$\int_0^2 (3x^2 - 1)dx = [x^3 - x]_0^2 = (8 - 2) - 0 = 6$

b $\int_2^0 (3x^2 - 1)dx$

$\int_2^0 (3x^2 - 1)dx = [x^3 - x]_2^0 = 0 - (8 - 2) = -6$

The integrals are opposites because the limits have been reversed.

c $\int_0^1 20x^3 dx$

$\int_0^1 20x^3 dx = [5x^4]_0^1 = 5$

d $20\int_0^1 x^3 dx$

$20\int_0^1 x^3 dx = 20\left[\frac{x^4}{4}\right]_0^1 = 20\left(\frac{1}{4}\right) = 5$

A constant factor can be taken out of the integral.

e $\int_1^4 (4x + 5)dx$

$\int_1^4 (4x + 5)dx = [2x^2 + 5x]_1^4$
$= (32 + 20) - (2 + 5)$
$= 45$

f $\int_1^4 4x\,dx$

$\int_1^4 4x\,dx = [2x^2]_1^4$
$= 32 - 2$
$= 30$

g $\int_1^4 5\,dx$

$\int_1^4 5\,dx = [5x]_1^4$
$= 20 - 5$
$= 15$

An integral of a sum is the sum of the integrals.

h $\int_0^2 12x^3 dx$

$\int_0^2 12x^3 dx = [3x^4]_0^2$
$= 48$

i $\int_0^1 12x^3 dx$

$\int_0^1 12x^3 dx = [3x^4]_0^1$
$= 3$

j $\int_1^2 12x^3 dx$

$\int_1^2 12x^3 dx = [3x^4]_1^2 = 48 - 3 = 45$

The interval [0, 2] can be split into [0, 1] and [1, 2].

8 Use the properties of the definite integral to evaluate each integral without using a primitive function. Give reasons.

a $\int_3^3 \sqrt{9 - x^2}\,dx$

0. The interval has zero width (both limits are 3).

b $\int_4^4 (x^3 - 3x^2 + 5x - 7)dx$

0. The interval has zero width (both limits are 4).

c $\int_{-1}^1 x^3 dx$

0. Integrand is an odd function and the interval [−1, 1] is symmetric about the origin.

d $\int_{-5}^5 (x^3 - 25x)dx$

0. Integrand is odd and the interval [−5, 5] is symmetric about the origin.

e $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin x\,dx$

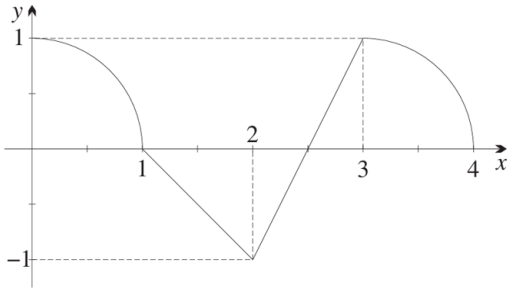
0. $f(x) = \sin x$ is an odd function and the interval $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ is symmetric about the origin.

f $\int_{-2}^2 \frac{x}{1 + x^2}dx$

0. Integrand is odd (numerator is odd, denominator is even) and the interval [−2, 2] is symmetric about the origin.

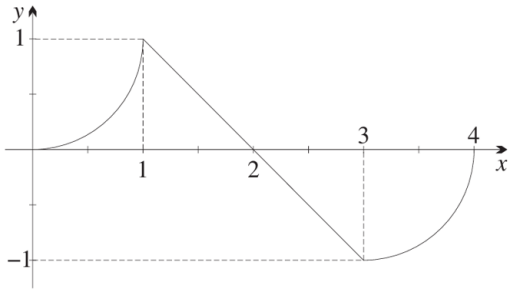
9 Use area formulae to find $\int_0^4 f(x)dx$, given the following sketches of $f(x)$.

a



$\frac{\pi}{2} - \frac{1}{2}$

b

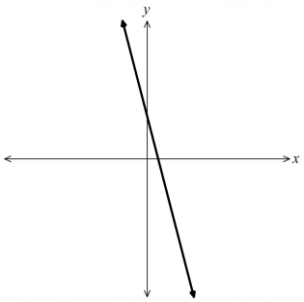


$1 - \frac{\pi}{2}$

10

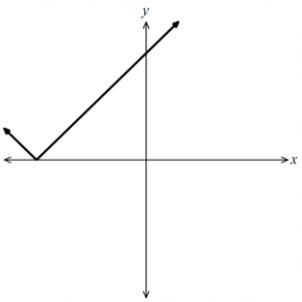
a Calculate these definite integrals using graphs and area formulae.

i $\int_{-\frac{1}{4}}^{\frac{1}{4}} (1 - 4x)dx$



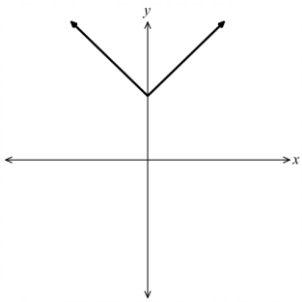
$\frac{1}{2}$

ii $\int_{-5}^{1} |x + 5|dx$



18

iii $\int_0^2 (|x| + 3)dx$



8

b Hence write down the value of:

i $\int_{\frac{1}{4}}^{-\frac{1}{4}} (1 - 4x)dx$

$-\frac{1}{2}$

ii $\int_1^{-5} |x + 5|dx$

-18

iii $\int_2^0 (|x| + 3)dx$

-8

Integrating Exponential Functions

Review

☐ Differentiate these functions.

a $f(x) = 4e^{x}$	**b** $f(x) = 4e^{3x}$	**c** $f(x) = 4e^{3x+1}$

Integrate exponential functions

$$f(x) = 3e^{2x+5}$$

Differentiate ↓ multiply by 2

$$f'(x) = 6e^{2x+5}$$

$$f(x) = \frac{6}{2}e^{2x+5} + C$$

Integrate ↑ divide by 2

$$f'(x) = 6e^{2x+5}$$

$\int 5e^{x}\,dx$	$\int 5e^{7x}\,dx$	$\int 3e^{7x}\,dx$	$\int 3e^{11x}\,dx$
$= 5e^{x} + C$	$= \frac{5}{7}e^{7x} + C$	$= \frac{3}{7}e^{7x} + C$	$= \frac{3}{11}e^{11x} + C$

Find the indefinite integral of the following:

a $\int 5e^{x}\,dx$	**b** $\int 5e^{7x}\,dx$	**c** $\int 3e^{7x}\,dx$
d $\int 3e^{11x}\,dx$	**e** $\int 3e^{11x-4}\,dx$	**f** $\int -\frac{e^{2-5x}}{7}\,dx$

Integrating Exponential Functions of Base e

$$\int e^x \, dx = e^x + C$$

$$\int e^{ax+b} \, dx = \frac{e^{ax+b}}{a} + C$$

Example Integrate exponential functions

$$\int e^{2x} + e^{-x} \, dx$$

$$= \frac{e^{2x}}{2} + \frac{e^{-x}}{-1} + C$$

$$= \frac{e^{2x}}{2} - \frac{1}{e^x} + C$$

$$\int_2^3 e^{5-2x} \, dx$$

$$= \left[-\frac{1}{2} e^{5-2x} \right]_2^3$$

$$= -\frac{1}{2}(e^{-1} - e^1)$$

$$= -\frac{1}{2}\left(\frac{1}{e} - e\right)$$

$$= \frac{e^2 - 1}{2e}$$

Guided Practice

a

$$\int \frac{e^x}{3} - e^{5x-2} \, dx$$

$\frac{e^x}{3} - \frac{e^{5x-2}}{5} + C$

b

$$\int_{-1}^3 4e^{\frac{x}{5}} \, dx$$

$20\left(e^{\frac{3}{5}} - e^{-\frac{1}{5}}\right)$

1 Find the indefinite integral.

a $\int e^{2x}\,dx$ $\left(\frac{1}{2}\right)e^{2x}+C$	**b** $\int e^{3x}\,dx$ $\left(\frac{1}{3}\right)e^{3x}+C$	**c** $\int e^{\frac{x}{3}}\,dx$ $3e^{\frac{x}{3}}+C$	**d** $\int e^{\frac{x}{2}}\,dx$ $2e^{\frac{x}{2}}+C$
e $\int 10e^{2x}\,dx$ $5e^{2x}+C$	**f** $\int 12e^{3x}\,dx$ $4e^{3x}+C$	**g** $\int e^{4x+5}\,dx$ $\left(\frac{1}{4}\right)e^{4x+5}+C$	**h** $\int e^{4x-2}\,dx$ $\left(\frac{1}{4}\right)e^{4x-2}+C$
i $\int 6e^{3x+2}\,dx$ $2e^{3x+2}+C$	**j** $\int 4e^{4x+3}\,dx$ $e^{4x+3}+C$	**k** $\int e^{7-2x}\,dx$ $-\left(\frac{1}{2}\right)e^{7-2x}+C$	**l** $\int \frac{1}{2}e^{1-3x}dx$ $-\left(\frac{1}{6}\right)e^{1-3x}+C$

2 Evaluate these definite integrals.

a $\int_0^1 e^x\,dx$ $e-1$	**b** $\int_1^2 e^x\,dx$ e^2-e	**c** $\int_{-1}^3 e^{-x}\,dx$ $e-e^{-3}$	**d** $\int_{-2}^0 e^{-x}\,dx$ e^2-1
e $\int_0^2 e^{2x}\,dx$ $\left(\frac{1}{2}\right)(e^4-1)$	**f** $\int_{-1}^2 20e^{-5x}\,dx$ $4(e^5-e^{-10})$	**g** $\int_{-3}^1 8e^{-4x}\,dx$ $2(e^{12}-e^{-4})$	**h** $\int_{-1}^3 9e^{6x}\,dx$ $\left(\frac{3}{2}\right)(e^{18}-e^{-6})$
i $\int_{-1}^1 e^{2x+1}\,dx$ $\left(\frac{1}{2}\right)(e^3-e^{-1})$	**j** $\int_{-2}^0 e^{4x-3}\,dx$ $\left(\frac{1}{4}\right)(e^{-3}-e^{-11})$	**k** $\int_{-2}^{-1} e^{3x+2}\,dx$ $\left(\frac{1}{3}\right)(e^{-1}-e^{-4})$	**l** $\int_{-\frac{1}{2}}^{\frac{1}{2}} e^{3-2x}\,dx$ $\left(\frac{e^2}{2}\right)(e^2-1)$
m $\int_{-\frac{1}{3}}^{\frac{1}{3}} e^{2+3x}\,dx$ $\left(\frac{e}{3}\right)(e^2-1)$	**n** $\int_1^2 6e^{3x+1}\,dx$ $2e^4(e^3-1)$	**o** $\int_2^3 12e^{4x-5}\,dx$ $3e^3(e^4-1)$	**p** $\int_1^2 12e^{8-3x}\,dx$ $4e^2(e^3-1)$

3 Find the indefinite integral of each function.

a $\int \frac{1}{e^x}\,dx$

$-e^{-x} + C$

b $\int \frac{1}{e^{2x}}\,dx$

$-\left(\frac{1}{2}\right)e^{-2x} + C$

c $\int \frac{1}{e^{3x}}\,dx$

$-\left(\frac{1}{3}\right)e^{-3x} + C$

d $-\int \frac{3}{e^{3x}}\,dx$

$e^{-3x} + C$

e $\int \frac{6}{e^{2x}}\,dx$

$-3e^{-2x} + C$

f $\int \frac{8}{e^{-2x}}\,dx$

$4e^{2x} + C$

4

a A function $f(x)$ has derivative $f'(x) = e^{2x}$.
Find the equation of $f(x)$, which will involve an arbitrary constant.

$\left(\frac{1}{2}\right)e^{2x} + C$

b It is also known that $f(0) = -2$.
Find the arbitrary constant and hence write down the equation of $f(x)$.

5

$f(x) = \left(\frac{1}{2}\right)e^{2x} - \frac{5}{2}$

a Find $f(1)$ and $f(2)$.

$f(1) = \left(\frac{1}{2}\right)e^{2} - \frac{5}{2}$

$f(2) = \left(\frac{1}{2}\right)e^{4} - \frac{5}{2}$

6 Find the indefinite integral of:

a $\int e^x(e^x + 1)\,dx$

$\left(\frac{1}{2}\right)e^{2x} + e^x + C$

b $\int e^{-x}(2e^{-x} - 1)\,dx$

$e^{-x} - e^{-2x} + C$

c $\int (e^x - 1)^2\,dx$

$\left(\frac{1}{2}\right)e^{2x} - 2e^x + x + C$

7 Find $f(x)$ and then find $f(1)$, given that:

a $f'(x) = 1 + 2e^x$ and $f(0) = 1$

$f(x) = x + 2e^x + C$

$f(0) = 1: 0 + 2 + C = 1, C = -1$

$f(x) = x + 2e^x - 1, f(1) = 1 + 2e - 1 = 2e$

b $f'(x) = e^{2x-1}$ and $f\left(\frac{1}{2}\right) = 3$

$f(x) = \left(\frac{1}{2}\right)e^{2x-1} + C$

$f\left(\frac{1}{2}\right) = 3: \left(\frac{1}{2}\right)e^0 + C = 3, C = \frac{5}{2}$

$f(x) = \left(\frac{1}{2}\right)e^{2x-1} + \frac{5}{2}, f(1) = \left(\frac{1}{2}\right)(e + 5)$

c $f'(x) = e^{\frac{x}{2}+1}$ and $f(-2) = -4$

$f(x) = 2e^{\frac{x}{2}+1} + C$

$f(-2) = -4: 2e^0 + C = -4, C = -6$

$f(x) = 2e^{\frac{x}{2}+1} - 6, f(1) = 2e^{\frac{3}{2}} - 6$

8 Find the indefinite integral of:

a $\int e^{7x+q}\,dx$	**b** $\int e^{3x-k}\,dx$	**c** $\int e^{sx+1}\,dx$	**d** $\int e^{kx-1}\,dx$
$\left(\frac{1}{7}\right)e^{7x+q} + C$	$\left(\frac{1}{3}\right)e^{3x-k} + C$	$\left(\frac{1}{s}\right)e^{sx+1} + C$	$\left(\frac{1}{k}\right)e^{kx-1} + C$
e $\int pe^{px+q}\,dx$	**f** $\int me^{mx+k}\,dx$	**g** $\int Ae^{sx-t}\,dx$	**h** $\int Be^{kx-\ell}\,dx$
$e^{px+q} + C$	$e^{mx+k} + C$	$\left(\frac{A}{s}\right)e^{sx-t} + C$	$\left(\frac{B}{k}\right)e^{kx-\ell} + C$

9 Find the indefinite integral of:

a $\int \frac{1}{e^{x-1}}\,dx$	**b** $\int \frac{1}{e^{3x-1}}\,dx$	**c** $\int \frac{1}{e^{2x+5}}\,dx$
$-e^{1-x} + C$	$-\left(\frac{1}{3}\right)e^{1-3x} + C$	$-\left(\frac{1}{2}\right)e^{-2x-5} + C$
d $\int \frac{4}{e^{2x-1}}\,dx$	**e** $\int \frac{10}{e^{2-5x}}\,dx$	**f** $\int \frac{12}{e^{3x-5}}\,dx$
$-2e^{1-2x} + C$	$2e^{5x-2} + C$	$-4e^{5-3x} + C$

10 By writing each integrand as the sum of multiples of powers of e, find:

a $\int \frac{e^x+1}{e^x}\,dx$

$x - e^{-x} + C$

b $\int \frac{e^{2x}+1}{e^x}\,dx$

$e^x - e^{-x} + C$

c $\int \frac{e^x-1}{e^{2x}}\,dx$

$\left(\frac{1}{2}\right)e^{-2x} - e^{-x} + C$

d $\int \frac{e^x-3}{e^{3x}}\,dx$

$e^{-3x} - \left(\frac{1}{2}\right)e^{-2x} + C$

e $\int \frac{2e^{2x}-3e^x}{e^{4x}}\,dx$

$e^{-3x} - e^{-2x} + C$

f $\int \frac{2e^x-e^{2x}}{e^{3x}}\,dx$

$e^{-x} - e^{-2x} + C$

11 Evaluate:

a $\int_0^1 e^x(2e^x - 1)\,dx$

$e^2 - e$

b $\int_{-1}^1 (e^x + 2)^2\,dx$

$\left(\frac{1}{2}\right)(e^2 - e^{-2}) + 4(e - e^{-1}) + 8$

c $\int_0^1 (e^x - 1)(e^{-x} + 1)\,dx$

$e + e^{-1} - 2$

d $\int_{-1}^1 (e^{2x} + e^{-x})(e^{2x} - e^{-x})\,dx$

$\left(\frac{1}{4}\right)(e^4 - e^{-4}) + \left(\frac{1}{2}\right)(e^{-2} - e^2)$

e $\int_0^1 \frac{e^{3x}+e^x}{e^{2x}}\,dx$

$e - e^{-1}$

f $\int_{-1}^1 \frac{e^x-1}{e^{2x}}\,dx$

$e - e^{-1} + \left(\frac{1}{2}\right)(e^{-2} - e^2)$

12 Write each integrand as a power of e, and hence find the indefinite integral.

a $\int \frac{1}{(e^x)^2}\,dx$

$-\left(\frac{1}{2}\right)e^{-2x} + C$

b $\int \frac{1}{(e^x)^3}\,dx$

$-\left(\frac{1}{3}\right)e^{-3x} + C$

c $\int \sqrt{e^x}\,dx$

$2e^{\frac{x}{2}} + C$

d $\int \sqrt[3]{e^x}\,dx$

$3e^{\frac{x}{3}} + C$

e $\int \frac{1}{\sqrt{e^x}}\,dx$

$-2e^{-\frac{x}{2}} + C$

f $\int \frac{1}{\sqrt[3]{e^x}}\,dx$

$-3e^{-\frac{x}{3}} + C$

13

a Find y as a function of x if $y' = e^{x-1}$, and $y = 1$ when $x = 1$.
What is the y-intercept of this curve?

$y = \int e^{x-1}\,dx = e^{x-1} + C$

$y(1) = 1: e^0 + C = 1$

$C = 0$

$y = e^{x-1}$

$y(0) = e^{-1}$

b The gradient of a curve is given by $y' = e^{2-x}$, and the curve passes through the point $(0, 1)$. What is the equation of this curve? What is its horizontal asymptote?

$y = \int e^{2-x}\,dx = -e^{2-x} + C$

$y(0) = 1: -e^2 + C = 1$

$C = e^2 + 1$

$y = e^2 + 1 - e^{2-x}$

Horizontal asymptote: as $x \to \infty$, $e^{2-x} \to 0$, so $y = e^2 + 1$

c It is known that $f'(x) = e^x + \frac{1}{e}$ and that $f(-1) = -1$. Find $f(0)$.

$f(x) = \int \left(e^x + \frac{1}{e}\right)dx = e^x + \frac{x}{e} + C$

$f(-1) = e^{-1} + \frac{-1}{e} + C = -1, C = -1$

$f(x) = e^x + \frac{x}{e} - 1$

$f(0) = 1 + 0 - 1 = 0$

d Given that $f''(x) = e^x - e^{-x}$ and that $y = f(x)$ is horizontal as it passes through the origin, find $f(x)$.

$f''(x) = e\text{^}x - e^{-x}$

$f'(x) = e^x + e^{-x} + C_1$

Horizontal at origin means $f'(0) = 0: 1 + 1 + C_1 = 0, C_1 = -2$

$f'(x) = e^x + e^{-x} - 2$

$f(x) = e^x - e^{-x} - 2x + C_2$

$f(0) = 0: 1 - 1 - 0 + C_2 = 0, C_2 = 0$

$f(x) = e^x - e^{-x} - 2x$

Integrating Reciprocal Functions

Review

☐ Differentiate these functions.

a	$f(x) = \ln(x)$	**b**	$f(x) = 2\ln(3x)$	**c**	$f(x) = \ln(5 - 3x)$	**d**	$f(x) = \ln(2x)$

Integrate reciprocal functions

$f(x) = 3\ln(2x + 5)$

↓ Differentiate

$f'(x) = \dfrac{6}{2x + 5}$

$f(x) = \dfrac{6}{2}\ln|2x + 5| + C$

↑ Integrate

$f'(x) = \dfrac{6}{2x + 5}$

$\int \frac{7}{x}dx$	$\int \frac{2}{x}dx$	$\int \frac{1}{2x}dx$	$\int \frac{11}{2x+5}dx$
$= 7\ln\|x\| + C$	$= 2\ln\|x\| + C$	$= \frac{1}{2}\ln\|x\| + C$ or $= \frac{1}{2}\ln\|2x\| + C$	$= \frac{11}{2}\ln\|2x + 5\| + C$

Find the indefinite integral of the following:

a $\int \frac{8}{x}dx$	**b** $\int \frac{5}{x}dx$	**c** $\int \frac{1}{5x}dx$
d $\int \frac{2}{5x+1}dx$	**e** $\int \frac{2}{1-5x}dx$	**f** $\int \frac{2}{(1-5x)^2}dx$

Integrating Reciprocal Functions

$$\int \frac{1}{x}\,dx = \ln|x| + C$$

$$\int \frac{1}{ax+b}\,dx = \frac{1}{a}\ln|ax+b| + C$$

⚠ We use absolute value signs because $\ln x$ is only defined for $x > 0$.
However, the graph of $\frac{1}{x}$ exists for negative x-values too, so areas under the curve still exist there. $\ln|x|$ ensures the formula works for both positive and negative values of x.

Example Integrate reciprocal functions

$$\int_1^e \frac{1}{x}\,dx$$

$$= \Big[\ln|x|\Big]_1^e$$

$$= \ln|e| - \ln|1|$$

$$= 1 - 0$$

$$= 1$$

$$\int \frac{1}{1-2x}\,dx$$

$$= -\frac{1}{2}\ln|1-2x| + C$$

Guided Practice

a

$$\int_3^5 \frac{1}{x-2}\,dx$$

$\ln 3$

b

$$\int \frac{3}{5x+7}\,dx$$

$\frac{3\ln|5x+7|}{5} + C$

c

$$\int_{-5}^{-3} \frac{1}{x}\,dx$$

$\ln 3 - \ln 5$

d

$$\int_e^{e^2} \frac{5}{x}\,dx$$

5

1 Integrate the following reciprocal functions.

a $\int \frac{2}{x}\,dx$

$2\ln|x| + C$

b $\int \frac{1}{3x}\,dx$

$\left(\frac{1}{3}\right)\ln|x| + C$

c $\int \frac{4}{5x}\,dx$

$\left(\frac{4}{5}\right)\ln|x| + C$

d $\int \frac{3}{2x}\,dx$

$\left(\frac{3}{2}\right)\ln|x| + C$

2 Find these indefinite integrals.

a $\int \frac{1}{4x+1}\,dx$

$\left(\frac{1}{4}\right)\ln|4x + 1| + C$

b $\int \frac{1}{5x-3}\,dx$

$\left(\frac{1}{5}\right)\ln|5x - 3| + C$

c $\int \frac{6}{3x+2}\,dx$

$2\ln|3x + 2| + C$

d $\int \frac{15}{5x+1}\,dx$

$3\ln|5x + 1| + C$

e $\int \frac{4}{4x+3}\,dx$

$\ln|4x + 3| + C$

f $\int \frac{dx}{3-x}$

$-\ln|3 - x| + C$

g $\int \frac{dx}{7-2x}$

$-\left(\frac{1}{2}\right)\ln|7 - 2x| + C$

h $\int \frac{4dx}{5x-1}$

$\left(\frac{4}{5}\right)\ln|5x - 1| + C$

i $12\int \frac{dx}{1-3x}$

$-4\ln|1 - 3x| + C$

3 Evaluate these definite integrals.

a $\int_1^5 \frac{1}{x}\,dx$

$\ln 5$

b $\int_1^3 \frac{1}{x}\,dx$

$\ln 3$

c $\int_{-8}^{-2} \frac{1}{x}\,dx$

$-2\ln 2$

d $\int_{-3}^{9} \frac{1}{x}\,dx$

undefined

e $\int_1^4 \frac{dx}{2x}$

$\ln 2$

f $\int_{-15}^{-5} \frac{dx}{5x}$

$-\left(\frac{1}{5}\right)\ln 3$

4 Evaluate these definite integrals to 4 s.f.

a $\int_0^1 \frac{dx}{x+1}$

0.6931

b $\int_{-7}^{-5} \frac{dx}{x+2}$

-0.5108

c $\int_{-5}^{-2} \frac{dx}{2x+3}$

-0.9730

d $\int_1^2 \frac{3}{5-2x}\,dx$

1.648

e $\int_{-1}^1 \frac{3}{7-3x}\,dx$

0.9163

f $\int_0^{11} \frac{5}{2x-11}\,dx$

undefined

5 Evaluate these definite integrals.

a $\int_1^e \frac{dx}{x}$

1

b $\int_1^{e^2} \frac{dx}{x}$

2

c $\int_e^{e^4} \frac{dx}{x}$

3

d $\int_{\sqrt{e}}^{e} \frac{dx}{x}$

$\frac{1}{2}$

6 Find the indefinite integral.

a $\int \frac{x+1}{x}\,dx$

$x + \ln|x| + C$

b $\int \frac{x+3}{5x}\,dx$

$\left(\frac{1}{5}\right)x + \left(\frac{3}{5}\right)\ln|x| + C$

c $\int \frac{1-8x}{9x}\,dx$

$\left(\frac{1}{9}\right)\ln|x| - \left(\frac{8}{9}\right)x + C$

d $\int \frac{3x^2-2x}{x^2}\,dx$

$3x - 2\ln|x| + C$

e $\int \frac{2x^2+x-4}{x}\,dx$

$x^2 + x - 4\ln|x| + C$

f $\int \frac{x^4-x+2}{x^2}\,dx$

$\left(\frac{1}{3}\right)x^3 - \ln|x| - \frac{2}{x} + C$

7 Find these indefinite integrals.

a $\int \frac{1}{3x-k}\,dx$

$\left(\frac{1}{3}\right)\ln|3x-k| + C$

b $\int \frac{1}{mx-2}\,dx$

$\left(\frac{1}{m}\right)\ln|mx-2| + C$

c $\int \frac{p}{px+q}\,dx$

$\ln|px+q| + C$

d $\int \frac{A}{sx-t}\,dx$

$\left(\frac{A}{s}\right)\ln|sx-t| + C$

8 Find $f(x)$, and then find $f(2)$, given that:

a $f'(x) = 1 + \frac{2}{x}$ and $f(1) = 1$

$f(x) = x + 2\ln|x| + C$

$f(1) = 1$:

$1 + 0 + C = 1$,

$C = 0$

$f(x) = x + 2\ln|x|$,

$f(2) = 2 + 2\ln 2$

b $f'(x) = 3 + \frac{5}{2x-1}$ and $f(1) = 0$

$f(x) = 3x + \left(\frac{5}{2}\right)\ln|2x - 1| + C$

$f(1) = 0$:

$3 + \left(\frac{5}{2}\right)\ln 1 + C = 0$

$C = -3$

$f(x) = 3x + \left(\frac{5}{2}\right)\ln|2x - 1| - 3$

$f(2) = 3 + \left(\frac{5}{2}\right)\ln 3$

c $f'(x) = 6x^2 + \frac{15}{3x+2}$ and $f(1) = 5\ln 5$

$f(x) = 2x^3 + 5\ln|3x + 2| + C$

$f(1) = 5\ln 5$:

$2 + 5\ln 5 + C = 5\ln 5$,

$C = -2$

$f(x) = 2x^3 + 5\ln|3x + 2| - 2$

$f(2) = 14 + 5\ln 8$

9 Find the value of a if:

a $\int_1^a \frac{1}{x}dx = 5$

$a = e^5$

b $\int_a^e \frac{1}{x}dx = 5$

$a = e^{-4}$

c $\int_a^{-1} \frac{1}{x}dx = -2$

$a = -e^2$ (since $a < -1$)

d $\int_{-e}^a \frac{1}{x}dx = -2$

$a = -e^{-1}$ (since a < 0, otherwise integral is undefined)

10

a Given that the derivative of $f(x)$ is $\frac{x^2+x+1}{x}$ and $f(1) = 1\frac{1}{2}$, find $f(x)$.

$\frac{x^2+x+1}{x} = x + 1 + \frac{1}{x}$

$f(x) = \left(\frac{1}{2}\right)x^2 + x + \ln|x| + C$

$f(1) = \frac{3}{2}$:

$\frac{1}{2} + 1 + 0 + C = \frac{3}{2}$

$C = 0$

$f(x) = x + \ln|x| + \left(\frac{1}{2}\right)x^2$

b Given that the derivative of $g(x)$ is $\frac{2x^3-3x-4}{x^2}$ and $g(2) = -3\ln 2$, find $g(x)$.

$\frac{2x^3-3x-4}{x^2} = 2x - \frac{3}{x} - 4x^{-2}$

$g(x) = x^2 - 3\ln|x| + \frac{4}{x} + C$

$g(2) = -3\ln 2$:

$4 - 3\ln 2 + 2 + C = -3\ln 2$

$C = -6$

$g(x) = x^2 - 3\ln|x| + \frac{4}{x} - 6$

11 Stella found the primitive of the function $\frac{1}{5x}$ by taking out a factor of $\frac{1}{5}$,

$$\int \frac{1}{5x}dx = \frac{1}{5}\int \frac{1}{x}dx = \frac{1}{5}\ln|x| + C_1$$

Bella used the standard form directly:

$$\int \frac{1}{5x}dx = \frac{1}{5}\ln|5x| + C_2,$$

Explain what is going on. Will this affect their result when finding a definite integral?

$\ln|5x| = \ln 5 + \ln|x|$, so $\ln|x|$ and $\ln|5x|$ differ only by a constant ln 5.

Thus $C_2 = C_1 - \left(\frac{1}{5}\right)\ln 5$.

For a definite integral, the constant cancels out when you do $F(b) - F(a)$.

Integrating Trigonometric Functions

Review

☐ Differentiate these functions.

a $f(x) = \sin(5x)$	**b** $f(x) = \cos(3x)$	**c** $f(x) = \tan(-6x)$
d $f(x) = \sin(5x + 3)$	**e** $f(x) = \cos(3x - 2)$	**f** $f(x) = \tan(-6x + 5)$

Integrating Trigonometric Functions

$$\int \sin x \, dx = -\cos x + C \qquad \int \sin(ax + b) \, dx = -\frac{1}{a}\cos(ax + b) + C$$

$$\int \cos x \, dx = \sin x + C \qquad \int \cos(ax + b) \, dx = \frac{1}{a}\sin(ax + b) + C$$

$$\int \sec^2 x \, dx = \tan x + C \qquad \int \sec^2(ax + b) \, dx = \frac{1}{a}\tan(ax + b) + C$$

⚠ For definite integrals of $\sec^2 x$, you cannot integrate across asymptotes at $\pm\frac{\pi}{2}$, $\pm\frac{3\pi}{2}$…

Integrate reciprocal functions

Find the indefinite integral of the following:

a $\int \sin(5x) \, dx$	**b** $\int \sin(5x + 3) \, dx$	**c** $\int 4\cos(5x + 3) \, dx$
d $\int \cos(-5x) \, dx$	**e** $\int \sec^2(-5x) \, dx$	**f** $\int \sec^2\left(-\frac{x}{5}\right) dx$

Example Integrate trigonometric functions

$$\int_0^{\frac{\pi}{6}} \cos(3x)\,dx$$

$$= \left[\frac{1}{3}\sin 3x\right]_0^{\frac{\pi}{6}}$$

$$= \frac{1}{3}\left(\sin\frac{\pi}{2} - \sin 0\right)$$

$$= \frac{1}{3}$$

$$\int_0^{\frac{\pi}{8}} \sec^2(2x+\pi)\,dx$$

$$= \left[\frac{1}{2}\tan(2x+\pi)\right]_0^{\frac{\pi}{8}}$$

$$= \frac{1}{2}\left(\tan\frac{5\pi}{4} - \tan\pi\right)$$

$$= \frac{1}{2}(1-0)$$

$$= \frac{1}{2}$$

Guided Practice

a

$$\int_0^{\pi} \cos x\,dx$$

0

b

$$\int_0^{\frac{\pi}{3}} 4\sec^2 x\,dx$$

$4\sqrt{3}$

c

$$\int_{\pi}^{2\pi} \sin\left(\frac{x}{4}\right)dx$$

$2\sqrt{2}$

d

$$\int_{\pi}^{2\pi} \cos\left(3x+\frac{\pi}{2}\right)dx$$

$\frac{2}{3}$

Integrating in Degrees

⚠ Integration requires the integral to use radian measure.
If the question gives degrees, you must convert to radians.

$$\int \cos x^\circ\,dx = \int \cos\left(\frac{\pi x}{180}\right)dx$$

1 Find these indefinite integrals.

a $\int \sec^2 x \, dx$

$\tan x + C$

b $\int \cos x \, dx$

$\sin x + C$

c $\int \sin x \, dx$

$-\cos x + C$

d $\int -\sin x \, dx$

$\cos x + C$

e $\int 2 \cos x \, dx$

$2 \sin x + C$

f $\int \cos 2x \, dx$

$\left(\frac{1}{2}\right) \sin 2x + C$

g $\int \left(\frac{1}{2}\right) \cos x \, dx$

$\left(\frac{1}{2}\right) \sin x + C$

h $\int \cos \left(\frac{x}{2}\right) dx$

$2 \sin \left(\frac{x}{2}\right) + C$

i $\int \sin 2x \, dx$

$-\left(\frac{1}{2}\right) \cos 2x + C$

j $\int \sec^2 5x \, dx$

$\left(\frac{1}{5}\right) \tan 5x + C$

k $\int \cos 3x \, dx$

$\left(\frac{1}{3}\right) \sin 3x + C$

l $\int \sec^2 \left(\frac{x}{3}\right) dx$

$3 \tan \left(\frac{x}{3}\right) + C$

m $\int \sin \left(\frac{x}{2}\right) dx$

$-2 \cos \left(\frac{x}{2}\right) + C$

n $\int -\cos \left(\frac{x}{5}\right) dx$

$-5 \sin \left(\frac{x}{5}\right) + C$

o $\int -4 \sin 2x \, dx$

$2 \cos 2x + C$

p $\int \left(\frac{1}{4}\right) \sin \left(\frac{x}{4}\right) dx$

$-\cos \left(\frac{x}{4}\right) + C$

q $\int 12 \sec^2 \left(\frac{x}{3}\right) dx$

$36 \tan \left(\frac{x}{3}\right) + C$

r $\int 2 \cos \left(\frac{x}{3}\right) dx$

$6 \sin \left(\frac{x}{3}\right) + C$

2 Evaluate:

a $\int_0^{\frac{\pi}{2}} \cos x \, dx$

1

b $\int_0^{\frac{\pi}{6}} \cos x \, dx$

$\frac{1}{2}$

c $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \sin x \, dx$

$\frac{1}{\sqrt{2}}$

d $\int_0^{\frac{\pi}{3}} \sec^2 x \, dx$

$\sqrt{3}$

e $\int_0^{\frac{\pi}{4}} 2 \cos 2x \, dx$

1

f $\int_0^{\frac{\pi}{3}} \sin 2x \, dx$

$\frac{3}{4}$

g $\int_0^{\frac{\pi}{2}} \sec^2 \left(\frac{x}{2}\right) dx$

2

h $\int_{\frac{\pi}{3}}^{\pi} \cos \left(\frac{x}{2}\right) dx$

1

i $\int_0^{\pi} (2 \sin x - \sin 2x) \, dx$

4

3

a The gradient function of a certain curve is given by $\frac{dy}{dx} = \sin x$.
If the curve passes through the origin, find its equation.

$y = -\cos x + C$
Passes through origin: $0 = -cos\ 0 + C = -1 + C, C = 1$
$y = 1 - \cos x$

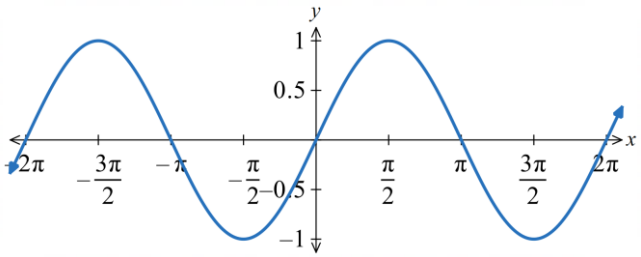
b Another curve passing through the origin has gradient function $y' = \cos x - 2\sin 2x$.
Find its equation.

$y = \sin x + \cos 2x + C$
$y(0) = 0: 0 + 1 + C = 0, C = -1$
$y = \sin x + \cos 2x - 1$

c If $\frac{dy}{dx} = \sin x + \cos x$, and $y = -2$ when $x = \pi$, find y as a function of x.

$y = -\cos x + \sin x + C$
$y(\pi) = -2: -(-1) + 0 + C = -2,$
$1 + C = -2$
$C = -3$
$y = -\cos x + \sin x - 3$

4 Evaluate these integrals, the graph of $y = \sin x$ is given.



a $\int_0^{\frac{\pi}{4}} \sin x\, dx$ $1 - \frac{1}{\sqrt{2}}$	**b** $\int_0^{\frac{\pi}{2}} \sin x\, dx$ 1	**c** $\int_0^{\frac{3\pi}{4}} \sin x\, dx$ $1 + \frac{1}{\sqrt{2}}$	**d** $\int_0^{\pi} \sin x\, dx$ 2
e $\int_0^{\frac{5\pi}{4}} \sin x\, dx$ $1 + \frac{1}{\sqrt{2}}$	**f** $\int_0^{\frac{3\pi}{2}} \sin x\, dx$ 1	**g** $\int_0^{\frac{7\pi}{4}} \sin x\, dx$ $1 - \frac{1}{\sqrt{2}}$	**h** $\int_0^{2\pi} \sin x\, dx$ 0

5 Find the following indefinite integrals.

a $\int \cos(x + 2)\,dx$ $\sin(x + 2) + C$	**b** $\int \cos(2x + 1)\,dx$ $\left(\frac{1}{2}\right)\sin(2x + 1) + C$	**c** $\int \sin(x + 2)\,dx$ $-\cos(x + 2) + C$
d $\int \sin(2x + 1)\,dx$ $-\left(\frac{1}{2}\right)\cos(2x + 1) + C$	**e** $\int \cos(3x - 2)\,dx$ $\left(\frac{1}{3}\right)\sin(3x - 2) + C$	**f** $\int \sin(7 - 5x)\,dx$ $\left(\frac{1}{5}\right)\cos(7 - 5x) + C$
g $\int \sec^2(4 - x)\,dx$ $-\tan(4 - x) + C$	**h** $\int \sec^2\left(\frac{1-x}{3}\right)dx$ $-3\tan\left(\frac{1 - x}{3}\right) + C$	**i** $\int \sin\left(\frac{1-x}{3}\right)dx$ $3\cos\left(\frac{1 - x}{3}\right) + C$

6

a Find $\int \left(6\cos 3x - 4\sin\left(\frac{x}{2}\right)\right)dx$.

$2\sin 3x + 8\cos\left(\frac{x}{2}\right) + C$

b Find $\int \left(8\sec^2 2x - 10\cos\left(\frac{x}{4}\right) - 36\sin\left(\frac{x}{3}\right)\right)dx$.

$4\tan(2x) + 108\cos\left(\frac{x}{3}\right) - 40\sin\left(\frac{x}{4}\right) + C$

7

a If $f'(x) = \pi\cos\pi x$ and $f(0) = 0$, find $f(x)$ and $f\left(\frac{1}{3}\right)$.

$f(x) = \int \pi\cos\pi x\,dx = \sin\pi x + C$

$f(0) = 0: \sin 0 + C = 0, C = 0$

$f(x) = \sin\pi x$

$f\left(\frac{1}{3}\right) = \sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}$

b If $f'(x) = \cos\pi x$ and $f(0) = \frac{1}{2\pi}$, find $f(x)$ and $f\left(\frac{1}{6}\right)$.

$f(x) = \int \cos\pi x\,dx = \left(\frac{1}{\pi}\right)\sin\pi x + C$

$f(0) = \frac{1}{2\pi}: 0 + C = \frac{1}{2\pi}, C = \frac{1}{2\pi}$

$f(x) = \frac{1}{2\pi} + \left(\frac{1}{\pi}\right)\sin\pi x$

$f\left(\frac{1}{6}\right) = \frac{1}{2\pi} + \left(\frac{1}{\pi}\right)\sin\left(\frac{\pi}{6}\right) = \frac{1}{\pi}$

8 If $f''(x) = 18\cos 3x$ and $f'(0) = f\left(\frac{\pi}{2}\right) = 1$, find $f(x)$.

$f''(x) = 18\cos 3x$

$f'(x) = 6\sin 3x + C_1$

$f'(0) = 1: 0 + C_1 = 1, C_1 = 1$

$f'(x) = 6\sin 3x + 1$

$f(x) = -2\cos 3x + x + C_2$

$f\left(\frac{\pi}{2}\right) = 1: -2\cos\left(\frac{3\pi}{2}\right) + \frac{\pi}{2} + C_2 = 1$

$-2(0) + \frac{\pi}{2} + C_2 = 1$

$C_2 = 1 - \frac{\pi}{2}$

$f(x) = -2\cos 3x + x + \left(1 - \frac{\pi}{2}\right)$

9 Find the following indefinite integrals, where a, b, u and v are constants.

a $\int a\sin(ax + b)\,dx$

$-\cos(ax + b) + C$

b $\int \pi^2\cos \pi x\,dx$

$\pi^2\left(\frac{1}{\pi}\right)\sin \pi x + C = \pi\sin \pi x + C$

c $\int \frac{1}{u}\sec^2(v + ux)\,dx$

$= \left(\frac{1}{u^2}\right)\tan(v + ux) + C$

d $\int \frac{a}{\cos^2 ax}\,dx$

$\tan ax + C$

10

a Using the Pythagorean identity, complete the statement: $1 + \tan^2 x = \ldots\ldots$,

$1 + \tan^2 x = \sec^2 x$

b Hence find $\int \tan^2 x\,dx$.

$\int \tan^2 x\,dx = \int(\sec^2 x - 1)\,dx = \tan x - x + C$

c Simplify $1 - \sin^2 x$, and hence find the value of $\int_0^{\frac{\pi}{3}} \frac{2}{1 - \sin^2 x}\,dx$.

$1 - \sin^2 x = \cos^2 x$

$\frac{2}{1 - \sin^2 x} = \frac{2}{\cos^2 x} = 2\sec^2 x$

$\int_0^{\frac{\pi}{3}} 2\sec^2 x\,dx = [2\tan x]_0^{\frac{\pi}{3}} = 2\sqrt{3}$

Integrating Exponentials of any Base

Review

☐ Differentiate these functions.

a $f(x) = 3^x$	**b** $f(x) = 5^{2x}$	**c** $f(x) = 7^{3x+2}$

Integrating Exponential Functions of any Base

$$\int a^x \, dx = \frac{a^x}{\ln a} + C$$

$$\int a^{kx} \, dx = \frac{a^{kx}}{k \ln a} + C$$

Integrate reciprocal functions

Find the indefinite integral of the following:

a $\int 3^x \, dx$	**b** $\int 3^{x+5} \, dx$	**c** $\int 3^{2x+5} \, dx$
d $\int 5^{2x+5} \, dx$	**e** $\int 5^{5-2x} \, dx$	**f** $\int 7^{\frac{x}{3}} \, dx$

1 Integrate:

a $\int 2^x \, dx$

$\frac{2^x}{\ln 2} + C$

b $\int 6^x \, dx$

$\frac{6^x}{\ln 6} + C$

c $\int 7^x \, dx$

$\frac{7^x}{\ln 7} + C$

d $\int 3^x \, dx$

$\frac{3^x}{\ln 3} + C$

2 Integrate:

a $\int_0^1 2^x \, dx$

1.443

b $\int_0^1 3^x \, dx$

1.820

c $\int_{-1}^1 5^x \, dx$

2.982

d $\int_0^2 4^x \, dx$

10.82

3 Give the exact value of each integral, then evaluate it correct to four decimal places.

a $\int_1^3 2^x \, dx$

$\frac{6}{\ln 2} \approx 8.6562$

b $\int_{-1}^1 (3^x + 1) \, dx$

$2 + \frac{8}{3 \ln 3} \approx 4.4273$

c $\int_0^2 (10^x - 10x) \, dx$

$\frac{99}{\ln 10} - 20 \approx 32.9952$

Mixed Practice

FOUNDATION

1 Work out the following:

a $\int \cos x \, dx$

$\sin x + C$

b $\int \sin x \, dx$

$-\cos x + C$

c $\int \sec^2 x \, dx$

$\tan x + C$

d $\int \frac{dx}{x^3}$

$-\frac{1}{2x^2} + C$

e $\int \frac{dx}{x^8}$

$-\frac{1}{7x^7} + C$

f $\int \frac{3}{x^2} dx$

$-\frac{3}{x} + C$

g $\int \frac{2}{x^5} dx$

$-\frac{1}{2x^4} + C$

h $\int \sqrt{x} \, dx$

$2\frac{\sqrt{x^3}}{3} + C$

i $\int x^{\frac{1}{3}} dx$

$\frac{3x^{\frac{4}{3}}}{4} + C$

j $\int \sqrt[3]{x} \, dx$

$3\frac{\sqrt[3]{x^4}}{4} + C$

k $\int x\sqrt{x} \, dx$

$2\frac{\sqrt{x^5}}{5} + C$

l $\int e^{4x} dx$

$\left(\frac{1}{4}\right)e^{4x} + C$

m $\int e^{-x} dx$

$-e^{-x} + C$

n $\int e^{5x} dx$

$\left(\frac{1}{5}\right)e^{5x} + C$

o $\int e^{-2x} dx$

$\left(-\frac{1}{2}\right)e^{-2x} + C$

p $\int \sin 3x \, dx$

$\left(-\frac{1}{3}\right)\cos 3x + C$

q $\int -\sin 7x \, dx$

$\left(\frac{1}{7}\right)\cos 7x + C$

r $\int \sec^2 5x \, dx$

$\left(\frac{1}{5}\right)\tan 5x + C$

s $\int \frac{2}{x} dx$

$2 \ln|x| + C$

t $\int \frac{5}{3x} dx$

$\left(\frac{5}{3}\right)\ln|x| + C$

u $\int \frac{1}{2x} dx$

$\left(\frac{1}{2}\right)\ln|x| + C$

v $\int (x - 2)(x + 5) \, dx$

$\frac{x^3}{3} + \frac{3x^2}{2} - 10x + C$

w $\int (1 - 2x)^2 \, dx$

$x - 2x^2 + \frac{4x^3}{3} + C$

x $\int (x^2 - x^{-7} + 5) \, dx$

$\frac{x^3}{3} + \frac{x^{-6}}{6} + 5x + C$

y $\int \sqrt{x}\left(1 + \frac{1}{\sqrt{x}}\right) dx$

$2\frac{\sqrt{x^3}}{3} + x + C$

z $\int (x + 1)^4 \, dx$

$\frac{(x + 1)^5}{5} + C$

aa $\int (3x - 4)^2 \, dx$

$\frac{(3x - 4)^3}{9} + C$

2 Work out the following:

a $\int (1 - x)^6 \, dx$

$-\frac{(1 - x)^7}{7} + C$

b $\int (5x - 1)^9 \, dx$

$\frac{(5x - 1)^{10}}{50} + C$

c $\int (3y - 2)^7 \, dy$

$\frac{(3y - 2)^8}{24} + C$

d $\int \cos (x + 1) \, dx$

$\sin (x + 1) + C$

e $\int \sin (2x - 3) \, dx$

$\left(-\frac{1}{2}\right) \cos (2x - 3) + C$

f $\int \cos (2x - 1) \, dx$

$\left(\frac{1}{2}\right) \sin (2x - 1) + C$

g $\int e^{4x+1} \, dx$

$\left(\frac{1}{4}\right) e^{4x+1} + C$

h $\int -3e^{5x} \, dx$

$\left(-\frac{3}{5}\right) e^{5x} + C$

i $\int (e^{7x} - 2) \, dx$

$\left(\frac{1}{7}\right) e^{7x} - 2x + C$

j $\int (e^{x-3} + x) \, dx$

$e^{x-3} + \frac{x^2}{2} + C$

k $\int \sin (\pi - x) \, dx$

$-\cos x + C$

l $\int \cos (x + \pi) \, dx$

$-\sin x + C$

m $\int 2 \sec^2 7x \, dx$

$\left(\frac{2}{7}\right) \tan 7x + C$

n $\int 4 \sin \left(\frac{x}{2}\right) dx$

$-8 \cos \left(\frac{x}{2}\right) + C$

o $\int 3 \sec^2 \left(\frac{x}{3}\right) dx$

$9 \tan \left(\frac{x}{3}\right) + C$

p $\int \frac{\sin x^\circ}{4} \, dx$

$-\left(\frac{45}{\pi}\right) \cos x^\circ + C$

q $\int (4 + 3x)^4 \, dx$

$\frac{(4 + 3x)^5}{15} + C$

r $\int (7x + 8)^{12} \, dx$

$\frac{(7x + 8)^{13}}{91} + C$

s $\int \sqrt{2x - 5} \, dx$

$\frac{\sqrt{(2x - 5)^3}}{3} + C$

t $\int \frac{2}{2x + 5} \, dx$

$\ln |2x + 5| + C$

u $\int \frac{4}{2x + 1} \, dx$

$2 \ln |2x + 1| + C$

v $\int \frac{x^6 - 3x^5 + 2x^4}{x^3} \, dx$

$\frac{x^4}{4} - x^3 + x^2 + C$

w $\int (x^2 - 4)(4x - 1) \, dx$

$x^4 - \frac{x^3}{3} - 8x^2 + 4x + C$

x $\int \frac{4x^3 - x^5 - 3x^2 + 7}{x^5} \, dx$

$-\frac{4}{x} + \left(\frac{3}{2x^2}\right) - \left(\frac{7}{4x^4}\right) - x + C$

3 Work out the following:

a $\int \frac{4}{4x-1}\,dx$

$\ln|4x-1|+C$

b $\int \frac{1}{x+3}\,dx$

$\ln|x+3|+C$

c $\int e^{2x}\,dx$

$\left(\frac{1}{2}\right)e^{2t}+C$

d $\int 3(x+7)^{-2}\,dx$

$-3(x+7)^{-1}+C$

e $\int \sqrt[3]{4x+}\,dx$

$\frac{3\sqrt[3]{(4x+3)^4}}{16}+C$

f $\int \sqrt{(t+3)^3}\,dt$

$2\frac{\sqrt{(t+3)^5}}{5}+C$

g $\int \sqrt{(5x+2)^5}\,dx$

$2\frac{\sqrt{(5x+2)^7}}{35}+C$

h $2\int(3x+1)^{-4}\,dx$

$-\frac{2(3x+1)^{-3}}{9}+C$

i $\int \frac{1}{2(4x-5)^3}\,dx$

$-\frac{1}{16(4x-5)^2}+C$

j $\int(2-x)^{-\frac{1}{2}}\,dx$

$-2(2-x)^{\frac{1}{2}}+C$

k $\int 5^x\,dx$

$\frac{5^x}{\ln 5}+C$

l $\int 7^{3x}\,dx$

$\frac{7^{3x}}{3\ln 7}+C$

m $\int 3^{2x-1}\,dx$

$\frac{3^{2x-1}}{2\ln 3}+C$

n $\int_0^{\frac{\pi}{2}} \cos x\,dx$

1

o $\int_0^1 e^{5x}\,dx$

$\left(\frac{1}{5}\right)(e^5-1)$

p $\int_1^2 (2x+1)^4\,dx$

$\frac{1441}{5}$

q $\int_0^1 (3y-2)^3\,dy$

$-\frac{5}{4}$

r $\int_1^2 (1-x)^7\,dx$

$-\frac{1}{8}$

s $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \sec^2 x\,dx$

$2\frac{\sqrt{3}}{3}$

t $\int_0^{\frac{\pi}{2}} \cos 3x\,dx$

$-\frac{1}{3}$

u $\int_0^2 -e^{-x}\,dx$

$\frac{1}{e^2}-1$

4 Work out the following:

a $\int_0^2 (3 - 2x)^5 \, dx$

$\frac{182}{3}$

b $\int_0^1 \frac{(3x-1)^2}{6} \, dx$

$\frac{1}{6}$

c $\int_4^5 (5 - x)^6 \, dx$

$\frac{1}{7}$

d $\int_3^6 \sqrt{x - 2} \, dx$

$\frac{14}{3}$

e $\int_0^{\frac{1}{2}} \sin(\pi x) \, dx$

$\frac{1}{\pi}$

f $\int_0^{\frac{\pi}{8}} \sec^2 2x \, dx$

$\frac{1}{2}$

g $\int_{\frac{\pi}{2}}^{\pi} \sin\left(\frac{x}{2}\right) dx$

$\sqrt{2}$

h $\int_0^{\frac{\pi}{12}} 3\cos 2x \, dx$

$\frac{3}{4}$

i $\int_0^{\frac{\pi}{10}} -\sin(5x) \, dx$

$-\frac{1}{5}$

j $\int_1^4 2e^{3x+4} \, dx$

$\left(\frac{2e^7}{3}\right)(e^9 - 1)$

k $\int_0^2 (e^{2x} + 1) \, dx$

$\left(\frac{1}{2}\right)e^4 + \frac{3}{2}$

l $\int_1^2 (e^x - x) \, dx$

$e^2 - e - \frac{3}{2}$

m $\int_2^3 (3x^2 - e^{2x}) \, dx$

$19 - \left(\frac{1}{2}\right)e^4(e^2 - 1)$

n $\int_0^3 (e^{2x} - e^{-x}) \, dx$

$\left(\frac{1}{2}\right)e^6 + e^{-3} - \frac{3}{2}$

o $\int_0^2 \frac{5}{(2x+1)^3} \, dx$

$\frac{6}{5}$

p $\int_1^4 \frac{2}{\sqrt{(5x-4)^3}} \, dx$

$\frac{3}{5}$

q $\int_1^3 \frac{2}{2x+5} \, dx$

0.5

r $\int_2^5 \frac{1}{x+1} \, dx$

0.7

s $\int \left(\cos x \cos\left(\frac{\pi}{3}\right) - \sin x \sin\left(\frac{\pi}{3}\right)\right) dx$

$\frac{\sin x + \sqrt{3}\cos x}{2} + C$

t $\int (\sin \pi \cos x - \cos \pi \sin x) \, dx$

$-\cos x + C$

5

a Show that $\frac{3x+3}{x^2-9} = \frac{1}{x+3} + \frac{2}{x-3}$

$$RHS = \frac{1}{x+3} + \frac{2}{x-3}$$
$$= \frac{(x-3)+2(x+3)}{(x+3)(x-3)}$$
$$= \frac{3x+3}{x^2-9} = LHS$$

b Hence find $\int \frac{3x+3}{x^2-9}\,dx$.

$$\ln|x+3| + 2\ln|x-3| + C$$

6

a Show that $\frac{x-6}{x-1} = 1 - \frac{5}{x-1}$.

$$RHS = 1 - \frac{5}{x-1}$$
$$= \frac{(x-1)-5}{x-1}$$
$$= \frac{x-6}{x-1} = LHS$$

b Hence find $\int \frac{x-6}{x-1}\,dx$.

$$x - 5\ln|x-1| + C$$

Reverse Chain Rule for Power Functions

Investigation Standard integral for power functions

a Differentiate $(x^3+5)^8$ using the chain rule.	**b** Therefore, find: $\int 24x^2(x^3+5)^7\,dx$	**c** and find: $\int x^2(x^3+5)^7\,dx$ $= \frac{1}{24}\int 24x^2(x^3+5)^7\,dx$
a Differentiate $\frac{1}{x^2+1}$ using the chain rule.	**b** Therefore, find: $\int -\frac{2x}{(1+x^2)^2}\,dx$	**c** and find: $\int \frac{x}{(1+x^2)^2}\,dx$

Reversing the chain rule requires you to identify the **inner function** and its **derivative** in the integrand.

You **can adjust for a missing constant**. Constants multiply every output by the same amount, so you can pull it outside the integral and recover the original function.

$$\int x^2(x^3+5)^7\,dx = \frac{1}{24}\int 24x^2(x^3+5)^3\,dx$$ ✓

You **cannot adjust for a missing variable term**. Variables change value depending on x, it does not scale every output equally and cannot be separated from the integral.

$$\int (x^3+5)^7\,dx = \frac{1}{24x^2}\int 24x^2(x^3+5)^7\,dx$$ ✗

So, you cannot use reverse chain rule to integrate $(x^3+5)^7$ because the integrand does not contain the derivative of the inner function, $3x^2$. The only way to integrate this expression is to expand it into individual terms.

Constants can move freely across an integral sign; variables cannot.

Identify whether you can integrate using reverse chain rule for power functions

We can integrate expressions of the form $f'(x)[f(x)]^n$, or any constant multiple of $f'(x)$.
This works because constants can be factored out of an integral, but variable terms cannot.

✗	✓	✓	✓	✗
$\int (x^2+5)^4\,dx$	$\int 2x(x^2+5)^4\,dx$	$\int x(x^2+5)^4\,dx$	$\int -7x(x^2+5)^4\,dx$	$\int x^3(x^2+5)^4\,dx$

Identify whether you can integrate the following:

a $\int 4x(2x^2+3)^5\,dx$	**b** $\int x^2(2x^2+3)^5\,dx$	**c** $\int x^4(x^5+1)^3 dx$	**d** $\int 6x(3x^2-4)^7\,dx$
e $\int x^3(x^5+1)^3\,dx$	**f** $\int 2x^2(x^3+2)^4\,dx$	**g** $\int (3x^2-4)^7 x\,dx$	**h** $\int x^2(1+x^5)^3\,dx$

Identify the adjustment factor for reverse chain rule of power functions

These expressions are of the form $f'(x)[f(x)]^n$. Adjust the following for integrating:

$\int x(x^2+3)^5 dx$	$\int x(3x^2+1)^4\,dx$	$\int 2x^2(x^3-5)^4\,dx$
$=\frac{1}{2}\int 2x(x^2+3)^5\,dx$	$=\frac{1}{6}\int 6x(3x^2+1)^4\,dx$	$=\frac{2}{3}\int 3x^2(x^3-5)^4\,dx$

a $\int x^3(x^4+1)^6\,dx$	**b** $\int x(2x^2+3)^5\,dx$	**c** $\int x^3(x^4+1)^{600}\,dx$	**d** $\int 2x^3(x^4+1)^{600}\,dx$
e $\int x(5x^2-3)^3\,dx$	**f** $\int 4x(5x^2-3)^3\,dx$	**g** $\int 4x^2(x^3+7)^8\,dx$	**h** $\int x^2(4x^3+7)^8\,dx$

ⓘ Reverse Chain Rule for Power Functions

Reverse chain rule requires the derivative of the inner function in the integrand.

$$\int f'(x)[f(x)]^n\,dx = \frac{[f(x)]^{n+1}}{n+1} + C$$

1. Identify the inner function $f(x)$.
2. Check whether the expression contains $f'(x)$.
   If it does not, you cannot use this rule.
3. Adjust for any missing constant factors.
4. Write the result as $\frac{[f(x)]^{n+1}}{n+1} + C$

Example Integrate using the reverse chain rule for power functions

$$\int x(x^2+1)^3\,dx$$

1. Inner function is $f(x) = x^2 + 1$
2. We need $f'(x) = 2x$
3. $$= \frac{1}{2}\int 2x(x^2+1)^3\,dx$$
4. $$= \frac{1}{2}\frac{(x^2+1)^4}{4} + C$$
$$= \frac{1}{8}(x^2+1)^4 + C$$

$$\int x^2\sqrt{x^3-1}\,dx$$

1. Inner function is $f(x) = x^3 - 1$
2. We need $f'(x) = 3x^2$
3. $$= \frac{1}{3}\int 3x^2(x^3-1)^{\frac{1}{2}}\,dx$$
4. $$= \frac{1}{3}\frac{(x^3-1)^{\frac{3}{2}}}{\left(\frac{2}{3}\right)} + C$$
$$= \frac{2}{9}\sqrt{(x^3-1)^3} + C$$

Guided Practice

a

$$\int x^2(x^3+5)^3\,dx$$

$\frac{1}{12}(x^3+5)^4 + C$

b

$$\int x\sqrt{1-x^2}\,dx$$

$-\frac{1}{3}\sqrt{(1-x^2)^3} + C$

1

a Find $\frac{d}{dx}(x^2 + 3)^4$

$8x(x^2 + 3)^3$

b $\int 8x(x^2 + 3)^3\,dx$

$(x^2 + 3)^4 + C$

c $\int 40x(x^2 + 3)^3\,dx$

$5(x^2 + 3)^4 + C$

2

a Find $\frac{d}{dx}(x^3 - 1)^5$

$15x^2(x^3 - 1)^4$

b $\int 15x^2(x^3 - 1)^4\,dx$

$(x^3 - 1)^5 + C$

c $\int 3x^2(x^3 - 1)^4\,dx$

$\frac{(x^3 - 1)^5}{5} + C$

3

a Find $\frac{d}{dx}\sqrt{2x^2 + 3}$

$\frac{2x}{\sqrt{2x^2 + 3}}$

b $\int \frac{2x}{\sqrt{2x^2+3}}\,dx$

$\sqrt{2x^2 + 3} + C$

c $\int \frac{x}{\sqrt{2x^2+3}}\,dx$

$\frac{1}{2}\sqrt{2x^2 + 3} + C$

4

a Find $\frac{d}{dx}(\sqrt{x} + 1)^3$

$\frac{3(\sqrt{x} + 1)^2}{2\sqrt{x}}$

b $\int \frac{3(\sqrt{x}+1)^2}{2\sqrt{x}}\,dx$

$(\sqrt{x} + 1)^3 + C$

c $\int \frac{(\sqrt{x}+1)^2}{\sqrt{x}}\,dx$

$\frac{2}{3}(\sqrt{x} + 1)^3 + C$

5

a Find $\frac{d}{dx}(x^3 + 3x^2 + 5)^4$

$12(x^2 + 2x)(x^3 + 3x^2 + 5)^3$

b $\int 12(x^2 + 2x)(x^3 + 3x^2 + 5)^3\,dx$

$(x^3 + 3x^2 + 5)^4 + C$

c $\int (x^2 + 2x)(x^3 + 3x^2 + 5)^3\,dx$

$1/12\ (x^3 + 3x^2 + 5)^4 + C$

6

a Find $\frac{d}{dx}(5 - x^2 - x)^7$

$-7(2x + 1)(5 - x^2 - x)^6$

b $\int(-14x - 7)(5 - x^2 - x)^6\,dx$

$(5 - x^2 - x)^7 + C$

c $\int(2x + 1)(5 - x^2 - x)^6\,dx$

$-\frac{1}{7}(5 - x^2 - x)^7 + C$

7 Integrate:

a $\int 5(5x + 4)^3\,dx$

$\frac{1}{4}(5x + 4)^4 + C$

b $\int -3(1 - 3x)^5\,dx$

$\frac{1}{6}(1 - 3x)^6 + C$

c $\int 2x(x^2 - 5)^7\,dx$

$\frac{1}{8}(x^2 - 5)^8 + C$

d $\int 3x^2(x^3 + 7)^4\,dx$

$\frac{1}{5}(x^3 + 7)^5 + C$

e $\int \frac{6x}{(3x^2+2)^2}\,dx$

$-\frac{1}{3x^2 + 2} + C$

f $\int -\frac{6x^2}{\sqrt{9 - 2x^3}}\,dx$

$2\sqrt{9 - 2x^3} + C$

8 Integrate:

a $\int 10x(5x^2 + 3)^2\,dx$

$\frac{1}{3}(5x^2 + 3)^3 + C$

b $\int 2x(x^2 + 1)^3\,dx$

$\frac{1}{4}(x^2 + 1)^4 + C$

c $\int 12x^2(1 + 4x^3)^5\,dx$

$\frac{1}{6}(1 + 4x^3)^6 + C$

d $\int x(1 + 3x^2)^4\,dx$

$\frac{1}{30}(1 + 3x^2)^5 + C$

e $\int x^3(1 - x^4)^7\,dx$

$-\frac{1}{32}(1 - x^4)^8 + C$

f $\int 3x^2\sqrt{x^3 - 1}\,dx$

$\frac{2}{3}(x^3 - 1)^{\frac{3}{2}} + C$

k $\int x\sqrt{5x^2 + 1}\,dx$

$\frac{1}{15}(5x^2 + 1)^{\frac{3}{2}} + C$

l $\int \frac{2x}{\sqrt{x^2+3}}\,dx$

$2\sqrt{x^2 + 3} + C$

m $\int \frac{x + 1}{\sqrt{4x^2+8x+1}}\,dx$

$\frac{1}{4}\sqrt{4x^2 + 8x + 1} + C$

9 Evaluate:

a $\int_{-1}^{1} x^2(x^3 + 1)^4\,dx$

$\frac{32}{15}$

b $\int_{0}^{1} \frac{x}{(5x^2+1)^3}\,dx$

$\frac{7}{144}$

c $\int_{0}^{\frac{1}{2}} x\sqrt{1 - 4x^2}\,dx$

$\frac{1}{12}$

d $\int_{-3}^{-1}(x + 5)(x^2 + 10x + 3)^2\,dx$

936

Further Reverse Chain Rule

Review

☐ Differentiate exponential functions using chain rule

a $\frac{d}{dx}\left(e^{x^2+3}\right)$ **b** $\frac{d}{dx}\left(e^{2-5x+x^3}\right)$ **c** $\frac{d}{dx}\left((1-e^{3x})^3\right)$

☐ Differentiate logarithmic functions using chain rule

a $\frac{d}{dx}\left(\ln(x^2+5)\right)$ **b** $\frac{d}{dx}\left(3\ln(9-x^3)\right)$ **c** $\frac{d}{dx}\left(\ln(e^x)\right)$

☐ Differentiate trigonometric functions using chain rule

a $\frac{d}{dx}\left(\sin(7x^2)\right)$ **b** $\frac{d}{dx}\left(\cos\left(\frac{x^2}{5}\right)\right)$ **c** $\frac{d}{dx}\left(\tan(4x^2-3)\right)$

ⓘ Reverse Chain Rule for Exponential Functions of Base e

$$\int f'(x)e^{f(x)}\,dx = e^{f(x)} + C$$

1. Identify the inner function $f(x)$ — the exponent of base e.
2. Check whether the expression contains $f'(x)$.
   If it does not, you cannot use this rule.
2. Adjust for any missing constant factors.
3. Write the result as $e^{f(x)} + C$

Example Integrate exponential functions using reverse chain rule

$$\int xe^{x^2+5}\,dx$$

1. Inner function is $x^2 + 5$
2. We need $2x$
3. $$= \frac{1}{2}\int 2xe^{x^2+5}\,dx$$
4. $$= \frac{1}{2}e^{x^2+5} + C$$

$$\int \frac{e^{2x}}{(1-e^{2x})^4}\,dx$$

1. Inner function is $1 - e^{2x}$
2. We need $-2e^{2x}$
3. $$= -\frac{1}{2}\int -2e^{2x}(1-e^{2x})^{-4}\,dx$$
4. $$= -\frac{1}{2}\frac{(1-e^{2x})^{-3}}{-3} + C$$
   $$= \frac{1}{6(1-e^{2x})^3} + C$$

Guided Practice

a

$$\int x^2e^{x^3+1}\,dx$$

$\frac{e^{x^3+1}}{3} + C$

b

$$\int \frac{e^{3x}}{(2+e^{3x})^5}\,dx$$

$-\frac{1}{12(e^{3x}+2)^4} + C$

c

$$\int \frac{e^{2x}}{(3+e^{2x})^3}\,dx$$

$-\frac{1}{4(e^{2x}+3)^2} + C$

d

$$\int xe^{1-x^2}\,dx$$

$-\frac{e^{1-x^2}}{2} + C$

Reverse Chain Rule for Reciprocal Functions

$$\int \frac{f'(x)}{f(x)}dx = \ln|f(x)| + C$$

1. Identify the function $f(x)$ — the function with power 1 in the denominator.
2. Check whether the numerator contains $f'(x)$.
   If it does not, you cannot use this rule.
3. Adjust for any missing constant factors.
4. Write the result as $\ln|f(x)| + C$.

Example Integrate reciprocal functions using reverse chain rule

$$\int \frac{2x}{x^2+2}dx$$

1. Inner function is $x^2 + 2$
2. We need $2x$
3. no adjustment needed
4. $= \ln|x^2 + 2| + C$

$$\int \frac{x}{9-x^2}dx$$

1. Inner function is $9 - x^2$
2. We need $-2x$
3. $= -\frac{1}{2}\int \frac{-2x}{9-x^2}dx$
4. $= -\frac{1}{2}\ln|9 - x^2| + C$

Guided Practice

a

$$\int \frac{3x^2}{x^3+5}dx$$

$\ln|x^3 + 5| + C$

b

$$\int \frac{6x}{3x^2+1}dx$$

$\ln|3x^2 + 1| + C$

c

$$\int \frac{x}{4-x^2}dx$$

$-\frac{\ln|4 - x^2|}{2} + C$

d

$$\int \frac{x}{5x^2-4}dx$$

$\frac{\ln|5x^2 - 4|}{10} + C$

Reverse Chain Rule for Trigonometric Functions

$$\int f'(x)\sin f(x)\,dx = -\cos f(x) + C$$

$$\int f'(x)\cos f(x)\,dx = \sin f(x) + C$$

$$\int f'(x)\sec^2 f(x)\,dx = \tan f(x) + C$$

Example Integrate trigonometric functions using reverse chain rule

$$\int x\sin(x^2)\,dx$$

1. Inner function is x^2
2. We need $2x$
3. $= \frac{1}{2}\int 2x\sin(x^2)\,dx$
4. $= -\frac{1}{2}\cos(x^2) + C$

$$\int x^2\sec^2(5x^3+2)\,dx$$

1. Inner function is $5x^3 + 2$
2. We need $15x^2$
3. $= \frac{1}{15}\int 15x^2\sec^2(5x^3+2)\,dx$
4. $= \frac{1}{15}\tan(5x^3+2) + C$

Guided Practice

a

$$\int 3x^2\cos(x^3)\,dx$$

$\sin(x^3)$

b

$$\int x^3\sec^2(x^4-1)\,dx$$

$\frac{\tan(x^4-1)}{4} + C$

c

$$\int x^3\sin(x^4-1)\,dx$$

$-\frac{\cos(x^4-1)}{4} + C$

d

$$\int \sin x\cos^2 x\,dx$$

$-\frac{\cos^3 x}{3} + C$

1

a Differentiate e^{x^2+3}.

$2xe^{x^2+3}$

b Hence find $\int 2xe^{x^2+3}\,dx$.

$e^{x^2+3} + C$

2

a Differentiate e^{x^2-2x+3}.

$2(x-1)e^{x^2-2x+3}$

b Hence find $\int (x-1)e^{x^2-2x+3}\,dx$.

$\frac{1}{2}e^{x^2-2x+3} + C$

3

a Differentiate e^{3x^2+4x+1}.

$(6x+4)e^{3x^2+4x+1}$

b Hence find $\int (3x+2)e^{3x^2+4x+1}\,dx$.

$\frac{1}{2}e^{3x^2+4x+1} + C$

4

a Differentiate $y = e^{x^3}$.

$3x^2e^{x^3}$

b Hence find $\int_{-1}^{0} x^2e^{x^3}\,dx$.

$\frac{1}{3}(1-e^{-1})$

5 Find the primitive:

a $\frac{2x}{x^2-9}$

$\ln|x^2-9| + C$

b $\frac{6x+1}{3x^2+x}$

$\ln|3x^2+x| + C$

c $\frac{2x+1}{x^2+x-3}$

$\ln|x^2+x-3| + C$

d $\frac{5-6x}{2+5x-3x^2}$

$\ln|2+5x-3x^2| + C$

e $\frac{x+3}{x^2+6x-1}$

$\frac{1}{2}\ln|x^2+6x-1| + C$

f $\frac{3-x}{12x-3-2x^2}$

$\frac{1}{4}\ln|12x-3-2x^2| + C$

g $\frac{e^x}{1+e^x}$

$\ln(1+e^x) + C$

h $\frac{e^{-x}}{1+e^{-x}}$

$-\ln(1+e^{-x}) + C$

i $\frac{e^x-e^{-x}}{e^x+e^{-x}}$

$\ln(e^x+e^{-x}) + C$

6 Find primitives of:

a xe^{x^2}

$\frac{1}{2}e^{x^2} + C$

b $4xe^{x^2-7}$

$2e^{x^2-7} + C$

c $(3x + 2)e^{3x^2+4x+1}$

$\frac{1}{2}e^{3x^2+4x+1} + C$

d $(x^2 - 2x)e^{x^3-3x^2}$

$\frac{1}{3}e^{x^3-3x^2} + C$

e $x^{-2}e^{x^{-1}}$

$-e^{x^{-1}} + C$

f $-\sqrt{x}e^{-x\sqrt{x}}$

$\frac{2}{3}e^{-x\sqrt{x}} + C$

7 Integrate:

a $\int \frac{3x^2}{x^3-5}\,dx$

$\int \frac{3x^2}{x^3-5}dx = \log_e|x^3 - 5| + C$

b $\int \frac{4x^3+1}{x^4+x-5}\,dx$

$\int \frac{4x^3+1}{x^4+x-5}dx = \log_e|x^4 + x - 5| + C$

c $\int \frac{x^3-3x}{x^4-6x^2}\,dx$

$\int \frac{x^3-3x}{x^4-6x^2}dx = \frac{1}{4}\log_e|x^4 - 6x^2| + C$

d $\int \frac{10x^3-7x}{5x^4-7x^2+8}\,dx$

$\int \frac{10x^3-7x}{5x^4-7x^2+8}dx$
$= \frac{1}{2}\log_e|5x^4 - 7x^2 + 8| + C$

e $\int_2^3 \frac{3x^2-1}{x^3-x}\,dx$

$\int_2^3 \frac{3x^2-1}{x^3-x}dx = [\log_e|x^3 - x|]_2^3$
$= \log_e 24 - \log_e 6$
$= \log_e 4 = 2\log_e 2$

f $\int_e^{2e} \frac{2x+2}{x^2+2x}\,dx$

$\int_e^{2e} \frac{2x+2}{x^2+2x}dx = [log_e|x^2 + 2x|]_e^{2e}$
$= log_e(4e^2 + 4e) - log_e(e^2 + 2e)$
$= log_e\left(\frac{4e(e+1)}{e(e+2)}\right)$
$= log_e\left(\frac{4(e+1)}{e+2}\right)$

8 Differentiate $e^{\sin x}$, and hence find $\int_0^{\frac{\pi}{2}} \cos x\, e^{\sin x}\,dx$.

$\frac{d}{dx}e^{\sin x} = \cos x\, e^{\sin x}$

$\int_0^{\frac{\pi}{2}} \cos x\, e^{\sin x}\,dx = [e^{\sin x}]_0^{\frac{\pi}{2}} = e - 1$

9 Differentiate $e^{\tan x}$, and hence find $\int_0^{\frac{\pi}{4}} \sec^2 x\, e^{\tan x}\, dx$.

$$\frac{d}{dx} e^{\tan x} = sec^2 x\, e^{\tan x}$$

$$\int_0^{\frac{\pi}{4}} sec^2 x\, e^{\tan x}\, dx = [e^{\tan x}]_0^{\frac{\pi}{4}} = e^1 - e^0 = e - 1$$

10 Find $\frac{d}{dx}(\sin^5 x)$, and hence find $\int \sin^4 x \cos x\, dx$.

$$\frac{d}{dx}(sin^5 x) = 5\, sin^4 x\, cos\, x$$

$$sin^4 x\, cos\, x = \frac{1}{5} \times 5\, sin^4 x\, cos\, x$$

$$\int sin^4 x\, cos\, x\, dx = \frac{1}{5} sin^5 x + C$$

11 Find $\frac{d}{dx}(\tan^3 x)$, and hence find $\int \tan^2 x \sec^2 x\, dx$.

$$\frac{d}{dx}(tan^3 x) = 3\, tan^2 x\, sec^2 x$$

$$tan^2 x\, sec^2 x = \frac{1}{3} \times 3\, tan^2 x\, sec^2 x$$

$$\int tan^2 x\, sec^2 x\, dx = \frac{1}{3} tan^3 x + C$$

12

a Show that $f(x) = xe^{-x^2}$ is an odd function.

$$f(-x) = (-x)e^{-(-x)^2}$$

$$= -xe^{-x^2}$$

$$= -f(x)$$

b Hence evaluate $\int_{-\sqrt{2}}^{\sqrt{2}} xe^{-x^2}\, dx$.

0

13 By treating the integrand as a reciprocal function, show that $\int_0^{\frac{\pi}{6}} \frac{\cos x}{1+\sin x}\, dx \approx 0.4$.

$$\int_0^{\frac{\pi}{6}} \frac{\cos x}{1 + \sin x} dx$$

$$f(x) = 1 + sin\, x, f'(x) = cos\, x$$

$$= [ln\,(1 + sin\, x)]_0^{\frac{\pi}{6}}$$

$$= \ln\left(1 + \frac{1}{2}\right) - \ln 1$$

$$= ln\left(\frac{3}{2}\right) \approx 0.4$$

14 Show that $\int \frac{e^x + 1}{e^{\frac{1}{2}x} + e^{-\frac{1}{2}x}}\,dx = 2e^{\frac{1}{2}x} + C$.

Multiply numerator and denominator by $e^{\frac{1}{2}x}$:

$$\frac{e^x + 1}{e^{\frac{1}{2}x} + e^{-\frac{1}{2}x}} = \frac{e^{\frac{1}{2}x}(e^x + 1)}{e^x + 1}$$

$$= e^{\frac{1}{2}x}$$

$$\int e^{\frac{1}{2}x}\,dx = 2e^{\frac{1}{2}x} + C$$

15 Use $\tan x = \frac{\sin x}{\cos x}$ to show that $\int_0^{\frac{\pi}{4}} \tan x\,dx = \frac{1}{2}\ln 2$.

$$\int_0^{\frac{\pi}{4}} \tan x\,dx = \int_0^{\frac{\pi}{4}} \frac{\sin x}{\cos x}dx$$

$$= [-\ln|\cos x|]_0^{\frac{\pi}{4}}$$

$$= -\ln\left(\frac{1}{\sqrt{2}}\right) + \ln 1$$

$$= \ln\sqrt{2}$$

$$= \frac{1}{2}\ln 2$$

16 Use $\cot x = \frac{\cos x}{\sin x}$ to find $\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \cot x\,dx$.

$$\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \frac{\cos x}{\sin x}\,dx = \left[\ln(\sin x)\right]_{\frac{\pi}{6}}^{\frac{\pi}{2}}$$

$$= \ln 1 - \ln\left(\frac{1}{2}\right) = \ln 2$$

17 Differentiate $(\log_e x)^2$, and hence find $\int_{\sqrt{e}}^{e} \frac{log_e x}{x}\,dx$.

$$\frac{d}{dx}(log_e x)^2 = 2\,log_e x \times \frac{1}{x} = 2\frac{log_e x}{x}$$

$$\frac{log_e x}{x} = \frac{1}{2} \times 2\frac{log_e x}{x}$$

$$\int_{\sqrt{e}}^{e} \frac{log_e x}{x}\,dx = \frac{1}{2}[(log_e x)^2]_{\sqrt{e}}^{e} = \frac{1}{2\left(1 - \frac{1}{4}\right)} = \frac{3}{8}$$

18 Differentiate $\ln(\ln x)$ and hence determine the family of primitives of $\frac{1}{x \ln x}$.

$$\frac{d}{dx}ln(ln\,x) = \frac{1}{ln\,x} \times \frac{1}{x} = \frac{1}{x\,ln\,x}$$

$$\int \frac{1}{x\,ln\,x}dx = ln(ln\,x) + C$$

19 Evaluate:

a $\int_0^{\pi} \sin x \cos^8 x \, dx$

$$\int_0^{\pi} \sin x \cos^8 x \, dx = \left[-\frac{\cos^9 x}{9}\right]_0^{\pi}$$
$$= -\frac{(-1)^9}{9} + \frac{1^9}{9} = \frac{2}{9}$$

b $\int_0^{\frac{\pi}{2}} \sin x \cos^n x \, dx$

$$\int_0^{\frac{\pi}{2}} \sin x \cos^n x \, dx = \left[-\frac{\cos^{n+1} x}{n+1}\right]_0^{\frac{\pi}{2}}$$
$$= 0 - \left(-\frac{1}{n+1}\right) = \frac{1}{n+1}$$

c $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos x \sin^7 x \, dx$

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos x \sin^7 x \, dx = \left[\frac{\sin^8 x}{8}\right]\left(-\frac{\pi}{2}\right)^{\frac{\pi}{2}}$$
$$= \frac{1}{8} - \frac{(-1)^8}{8}$$
$$= 0$$

d $\int_0^{\frac{\pi}{6}} \cos x \sin^n x \, dx$

$$\int_0^{\frac{\pi}{6}} \cos x \sin^n x \, dx = \left[\frac{\sin^{n+1} x}{n+1}\right]_0^{\frac{\pi}{6}}$$
$$= \frac{\left(\frac{1}{2}\right)^{n+1}}{n+1}$$
$$= \frac{1}{2^{(n+1)(n+1)}}$$

e $\int_0^{\frac{\pi}{3}} \sec^2 x \tan^7 x \, dx$

$$\int_0^{\frac{\pi}{3}} \sec^2 x \tan^7 x \, dx = \left[\frac{\tan^8 x}{8}\right]_0^{\frac{\pi}{3}}$$
$$= \frac{(\sqrt{3})^8}{8}$$
$$= 10\frac{1}{8}$$

f $\int_0^{\frac{\pi}{4}} \sec^2 x \tan^n x \, dx$

$$\int_0^{\frac{\pi}{4}} \sec^2 x \tan^n x \, dx = \left[\frac{\tan^{n+1} x}{n+1}\right]_0^{\frac{\pi}{4}}$$
$$= \frac{1^{n+1}}{n+1}$$
$$= \frac{1}{n+1}$$

20

a Differentiate $y = x \log_e x - x$.

b Hence find $\int \log_e x \, dx$ and $\int_{\sqrt{e}}^{e} \log_e x \, dx$.

$$y = x \log_e x - x$$
$$y' = \log_e x + x\left(\frac{1}{x}\right) - 1 = \log_e x$$

$$\int \log_e x \, dx = x \log_e x - x + C$$

$$\int \sqrt{e}^{e} \log_e x \, dx = [x \log_e x - x]\sqrt{e}^{e} = (e - e) - \left(\sqrt{e} \times \frac{1}{2} - \sqrt{e}\right) = 0 - \left(-\frac{\sqrt{e}}{2}\right) = \frac{\sqrt{e}}{2}$$

21

a Differentiate $y = xe^x$, and hence find $\int_0^2 xe^x\,dx$.

$y' = e^x + xe^x$

$xe^x = y' - e^x$

$\therefore \int_0^2 xe^x\,dx = \int_0^2 (y' - e^x)\,dx$

$= [y - e^x]_0^2$

$= [xe^x - e^x]_0^2$

$= e^2 + 1$

b Differentiate $y = xe^{-x}$, and hence find $\int_{-2}^0 xe^{-x}\,dx$.

$y' = e^{-x} - xe^{-x}$

$xe^{-x} = -y' + e^{-x}$

$\int_{-2}^0 xe^{-x}\,dx = \int_{-2}^0 -y' + e^{-x}\,dx$

$= [-y - e^{-x}]_{-2}^0$

$= [xe^{-x} - e^{-x}]_{-2}^0$

$= -1 - e^2$

22

a Find $\int \frac{e^x}{e^x + 1}\,dx$

$\int \frac{e^x}{e^x + 1}dx = log_e(e^x + 1) + C$

b Find $\int_1^e \left(x + \frac{1}{x^2}\right)^2 dx$

Expanding: $\left(x + \frac{1}{x^2}\right)^2 = x^2 + \frac{2}{x} + x^{-4}$

$\int_1^e \left(x^2 + \frac{2}{x} + x^{-4}\right)dx = \left[\frac{x^3}{3} + 2\,log_e x - \frac{x^{-3}}{3}\right]_1^e$

$= \left(\frac{e^3}{3} + 2 - \frac{1}{3e^3}\right) - \left(\frac{1}{3} + 0 - \frac{1}{3}\right)$

$= \frac{e^3}{3} + 2 - \frac{1}{3e^3} - 0$

$= \frac{1}{3}(e^3 - e^{-3}) + 2$

23 Show that $\frac{d}{dx}(\sin x - x\cos x) = x\sin x$, and hence find $\int_0^{\frac{\pi}{2}} x\sin x\,dx$.

$\frac{d}{dx}(sin\,x - x\,cos\,x) = \cos x - (\cos x - x\sin x)$

$= \cos x - \cos x + x\sin x$

$= x\,sin\,x$

$\int_0^{\frac{\pi}{2}} x\,sin\,x\,dx = [sin\,x - x\,cos\,x]_0^{\frac{\pi}{2}}$

$= (1 - 0) - (0 - 0)$

$= 1$

24

a Find the derivative of $y = 2x^2\log_e x - x^2$.

b Hence find $\int x \log_e x \, dx$ and $\int_e^2 x \log_e x \, dx$.

$y = 2x^2 log_e x - x^2$

$y' = 4x \, log_e x + 2x^2\left(\frac{1}{x}\right) - 2x$

$= 4x \, log_e x + 2x - 2x$

$= 4x \, log_e x$

$\int x \, log_e x \, dx = \frac{1}{4(2x^2 log_e x - x^2)} + C = \frac{1}{2}x^2 log_e x - \frac{1}{4}x^2 + C$

$\int_e^2 x \, log_e x \, dx = \left[\frac{1}{2}x^2 log_e x - \frac{1}{4}x^2\right]_e^2$

$= \left(\frac{1}{2(4)log_e 2} - 1\right) - \left(\frac{1}{2}e^2 - \frac{1}{4}e^2\right)$

$= 2 \, log_e 2 - 1 - \frac{e^2}{4}$

25

a Show that you can integrate the following in two different ways:

1. treating the integrand as $f'(x)f(x)$ and
2. Using $\sin 2x = 2 \sin x \cos x$

$$\int \sin x \cos x \, dx = \frac{1}{2}\sin^2 x + C = -\frac{1}{4}\cos 2x + D$$

Method 1: Let $f(x) = \sin x, f'(x) = \cos x$

$\int \sin x \cos x \, dx = \frac{\sin^2 x}{2} + C$

Method 2: Using $\sin 2x = 2 \sin x \cos x$, so $\sin x \cos x = \frac{1}{2}\sin 2x$

$\int \frac{1}{2}\sin 2x \, dx = -\frac{1}{4}\cos 2x + D$

b Show that $\frac{1}{2}\sin^2 x + C = -\frac{1}{4}\cos 2x + D$

$\sin^2 x = \frac{1}{2}(1 - \cos 2x)$,

so $\frac{1}{2}\sin^2 x + C = \frac{1}{4} - \frac{1}{4}\cos 2x + C$

$= -\frac{1}{4}\cos 2x + \left(C + \frac{1}{4}\right)$

So $D = C + \frac{1}{4}$. The two answers differ by a constant.

26 Find $\frac{d}{dx}(x \sin 2x)$, and hence find $\int_0^{\frac{\pi}{4}} x \cos 2x \, dx$.

$$\frac{d}{dx}(x \sin 2x) = \sin 2x + 2x \cos 2x$$

$$\text{Rearranging: } x \cos 2x = \frac{1}{2}\left(\frac{d}{dx}(x \sin 2x) - \sin 2x\right)$$

$$\int_0^{\frac{\pi}{4}} x \cos 2x \, dx = \frac{1}{2}\left[x \sin 2x + \frac{1}{2}\cos 2x\right]_0^{\frac{\pi}{4}}$$

$$= \frac{1}{2}\left(\left(\frac{\pi}{4}\right)(1) + \frac{1}{2(0)}\right) - \frac{1}{2\left(0 + \frac{1}{2}\right)} = \frac{\pi - 2}{8}$$

27 Show that $\frac{d}{dx}\left(\frac{1}{3}\cos^3 x - \cos x\right) = \sin^3 x$, and hence find $\int_0^{\frac{\pi}{3}} \sin^3 x \, dx$.

$$\frac{d}{dx}\left(\frac{1}{3}\cos^3 x - \cos x\right) = -\sin x \cos^2 x + \sin x = \sin x(1 - \cos^2 x) = \sin^3 x$$

$$\int_0^{\frac{\pi}{3}} \sin^3 x \, dx = \left[\frac{1}{3}\cos^3 x - \cos x\right]_0^{\frac{\pi}{3}}$$

$$= \left(\frac{1}{3}\left(\frac{1}{2}\right)^3 - \frac{1}{2}\right) - \left(\frac{1}{3} - 1\right) = \frac{5}{24}$$

28

a Show that $\frac{d}{dx}(\tan^3 x) = 3(\sec^4 x - \sec^2 x)$.

b Hence find $\int_0^{\frac{\pi}{4}} \sec^4 x \, dx$.

$$\frac{d}{dx}(\tan^3 x) = 3 \tan^2 x \sec^2 x = 3 \tan^2 x \sec^2 x$$

$$\text{Using } \tan^2 x = \sec^2 x - 1:$$

$$= 3(\sec^2 x - 1)\sec^2 x = 3(\sec^4 x - \sec^2 x)$$

$$\int_0^{\frac{\pi}{4}} \sec^4 x \, dx = \left[\frac{1}{3}\tan^3 x + \tan x\right]_0^{\frac{\pi}{4}} = \frac{1}{3(1)} + 1 - 0 = \frac{4}{3}$$

Mixed Practice

ⓘ Standard Integrals

$$\int x^n \, dx = \frac{1}{n+1} x^{n+1} + C$$

$$\int e^x \, dx = e^x + C$$

$$\int a^x \, dx = \frac{a^x}{\ln a} + C$$

$$\int \frac{1}{x} dx = \ln|x| + C$$

$$\int \sin x \, dx = -\cos x + C$$

$$\int \cos x \, dx = \sin x + C$$

$$\int \sec^2 x \, dx = \tan x + C$$

$$\int f'(x)[f(x)]^n \, dx = \frac{1}{n+1} f(x)^{n+1} + C$$

$$\int f'(x) e^{f(x)} \, dx = e^{f(x)} + C$$

$$\int f'(x) a^{f(x)} \, dx = \frac{a^{f(x)}}{\ln a} + C$$

$$\int \frac{f'(x)}{f(x)} dx = \ln|f(x)| + C$$

$$\int f'(x) \sin f(x) \, dx = -\cos f(x) + C$$

$$\int f'(x) \cos f(x) \, dx = \sin f(x) + C$$

$$\int f'(x) \sec^2 f(x) \, dx = \tan f(x) + C$$

Example Integrate functions

$$\int \cos^3 x \sin x \, dx$$

$$= \int \sin x \, (\cos x)^3 \, dx$$

1. Inner function is $\cos x$
2. We need $-\sin x$
3. $$= -\int -\sin x \, (\cos x)^3 \, dx$$
4. $$= -\frac{\cos^4 x}{4} + C$$

$$\int \frac{\cos(2x)}{\sin(2x)} dx$$

1. Inner function is $\sin 2x$
2. We need $2\cos 2x$
3. $$= \frac{1}{2}\int \frac{2\cos 2x}{\sin 2x} dx$$
4. $$= \frac{1}{2}\ln|\sin 2x| + C$$

$$\int 5\sec^2(3x)\, e^{\tan(3x)} \, dx$$

1. Inner function is $\tan 3x$
2. We need $3\sec^2 3x$
3. $$= \frac{5}{3}\int 3\sec^2(3x)\, e^{\tan 3x} \, dx$$
4. $$= \frac{5}{3} e^{\tan 3x} + C$$

$$\int \frac{xe^{x^2}}{\left(e^{x^2}+1\right)^3} dx$$

$$= \int xe^{x^2}\left(e^{x^2}+1\right)^{-3} dx$$

1. Inner function is $e^{x^2}+1$
2. We need $2xe^{x^2}$
3. $$= \frac{1}{2}\int 2xe^{x^2}\left(e^{x^2}+1\right)^{-3} dx$$
4. $$= \frac{1}{2}\frac{\left(e^{x^2}+1\right)^{-2}}{-2} + C$$
$$= -\frac{1}{4\left(e^{x^2}+1\right)^2} + C$$

1 Integrate:

a $\int 2x(x^2 + 1)^5 \, dx$

$\frac{(x^2 + 1)^6}{6} + C$

b $\int \frac{e^x}{e^x + 3} dx$

$\ln|e^x + 3| + C$

c $\int 3x^2(x^3 + 1)^3 \, dx$

$\frac{(x^3 + 1)^4}{4} + C$

d $\int \frac{\cos x}{2 + \sin x} \, dx$

$\ln|2 + \sin x| + C$

e $\int \cos x \cdot e^{\sin x} \, dx$

$e^{\sin x} + C$

f $\int 4x^3(x^4 + 5)^2 \, dx$

$\frac{(x^4 + 5)^3}{3} + C$

g $\int \frac{5x^4}{x^5 - 2} \, dx$

$\ln|x^5 - 2| + C$

h $\int \cos x \, (\sin x)^3 \, dx$

$\frac{\sin^4 x}{4} + C$

i $\int \frac{3x^2}{x^3 - 7} \, dx$

$\ln|x^3 - 7| + C$

j $\int \sec^2 x \cdot e^{\tan x} \, dx$

e^(tan x) + C

k $\int \frac{x}{x^2 + 4} \, dx$

$\left(\frac{1}{2}\right)\ln|x^2 + 4| + C$

l $\int x \cdot e^{x^2} \, dx$

$\frac{e^{x^2}}{2} + C$

m $\int (2x + 3)(x^2 + 3x - 2)^4 \, dx$

$\frac{(x^2 + 3x - 2)^5}{5} + C$

n $\int \frac{\sec^2 x}{1 + \tan x} \, dx$

$\ln|1 + \tan x| + C$

o $\int \left(\frac{1}{x}\right) \cdot e^{\ln x} \, dx$

$x + C$

p $\int x(3x^2 - 7)^6 \, dx$

$\frac{(3x^2 - 7)^7}{42} + C$

q $\int \frac{x+1}{x^2+2x-5}\,dx$

$\left(\frac{1}{2}\right)\ln|x^2+2x-5|+C$

r $\int e^x(e^x+1)^6\,dx$

$\frac{(e^x+1)^7}{7}+C$

s $\int \sec^2 x \tan^3 x\,dx$

$\frac{\tan^4 x}{4}+C$

t $\int \frac{(\ln x)^2}{x}\,dx$

$\frac{(\ln x)^3}{3}+C$

u $\int x\sqrt{x^2-5}\,dx$

$\frac{(x^2-5)^{\frac{3}{2}}}{3}+C$

v $\int \frac{4x}{2x^2+1}\,dx$

$\ln|2x^2+1|+C$

w $\int x^2(4-5x^3)^2\,dx$

$-\frac{(4-5x^3)^3}{45}+C$

x $\int \cos x \cdot (1+\sin x)^{-2}\,dx$

$-\frac{1}{1+\sin x}+C$

y $\int \frac{\sin x}{\cos^3 x}\,dx$

$\frac{\sec^2 x}{2}+C$

z $\int \frac{x^5}{2x^6+5}\,dx$

$\left(\frac{1}{12}\right)\ln|2x^6+5|+C$

aa $\int 3x(5x^2+3)^7\,dx$

$\frac{3(5x^2+3)^8}{80}+C$

bb $\int \frac{x+3}{x^2+6x+2}\,dx$

$\left(\frac{1}{2}\right)\ln|x^2+6x+2|+C$

cc $\int 4x^5(2x^6-3)^4\,dx$

$\frac{(2x^6-3)^5}{15}+C$

dd $\int \frac{1}{x^2}\,dx$

$-\frac{1}{x}+C$

ee $\int (x+2)(x^2+4x)^5\,dx$

$\frac{(x^2+4x)^6}{12}+C$

ff $\int \frac{x^2}{2x^3-7}\,dx$

$\left(\frac{1}{6}\right)\ln|2x^3-7|+C$

gg $\int (3x^2 - 2)(3x^3 - 6x - 2)^3 \, dx$

$\frac{(3x^3 - 6x - 2)^4}{12} + C$

hh $\int 2x \, e^{1-x^2} \, dx$

$-e^{1-x^2} + C$

ii $\int \sin x \cdot e^{\cos x} \, dx$

$-e^{\cos x} + C$

jj $\int \frac{e^{2x}}{e^{2x} + 5} \, dx$

$\left(\frac{1}{2}\right) \ln |e^{2x} + 5| + C$

kk $\int x^2 (x^3 - 1)^{-\frac{3}{2}} \, dx$

$-\frac{2}{3\sqrt{x^3 - 1}} + C$

ll $\tan x \, dx$

$-\ln |\cos x| + C$

mm $\int_0^2 x(2x^2 + 3)^2 \, dx$

$\frac{326}{3}$

nn $\int_0^1 x^2 (x^3 - 1)^5 \, dx$

$-\frac{1}{18}$

oo $\int_1^7 \frac{x^2}{x^3 + 2} \, dx$

$\left(\frac{1}{3}\right) \ln 115$

pp $\int_0^3 \frac{4x + 1}{2x^2 + x + 1} \, dx$

$\ln 22$

qq $\int_1^2 x^4 (x^5 + 2)^3 \, dx$

66 812.75

rr $\int_0^{\frac{\pi}{2}} \cos x \cdot e^{\sin x} \, dx$

$e - 1$

2

a Differentiate $x \ln x - x$, and hence find $\int \ln x \, dx$.

$$\frac{d}{dx}(x \ln x - x) = \ln x + 1 - 1 = \ln x$$

$$\int \ln x \, dx = x \ln x - x + C$$

b Use the change-of-base formula and part **a** to evaluate $\int_1^{10} \log_{10} x \, dx$.

$$\log_{10} x = \frac{\ln x}{\ln 10}$$

$$\int_1^{10} \log_{10} x \, dx = \left(\frac{1}{\ln 10}\right)[x \ln x - x]_1^{10}$$

$$= \left(\frac{1}{\ln 10}\right)[(10 \ln 10 - 10) - (0 - 1)]$$

$$= \left(\frac{1}{\ln 10}\right)(10 \ln 10 - 9) = 10 - \frac{9}{\ln 10}$$

3 Differentiate $x^2 e^x$. Hence find $\int x(2 + x)e^x \, dx$.

$$\frac{d}{dx}(x^2 e^x) = 2xe^x + x^2 e^x = x(2 + x)e^x$$

$$\text{Therefore } \int x(2 + x)e^x \, dx = x^2 e^x + C$$

4 Integrate:

a $\int 3^{5x} \, dx$

$$\frac{3^{5x}}{5 \log_e 3} + C$$

b $\int 6^{2x+7} \, dx$

$$\frac{6^{2x+7}}{2 \log_e 6} + C$$

c $\int 5 \times 7^{4-9x} \, dx$

$$-5 \times \frac{7^{4-9x}}{9 \log_e 7} + C$$

5 A function has $\frac{dy}{dx} = x^2(x^3 - 2)^4$ and passes through the point $(1, 4)$. Find its equation.

$$\frac{dy}{dx} = x^2(x^3 - 2)^4$$

$$y = \frac{\left(\frac{1}{3}\right)(x^3 - 2)^5}{5} + C = \frac{(x^3 - 2)^5}{15} + C$$

$$\text{Sub } (1, 4): 4 = \frac{(-1)^5}{15} + C = -\frac{1}{15} + C$$

$$C = \frac{61}{15}$$

$$y = \left(\frac{1}{15}\right)((x^3 - 2)^5 + 61)$$

6 A function has $f'(x) = x^2e^{2x^3}$ and passes through the point $(0, 0)$.
Find the equation of the function.

$$f'(x) = x^2e^{2x^3}$$
$$f(x) = \left(\frac{1}{6}\right)e^{2x^3} + C$$
$$\text{Sub } (0,0)\text{: } 0 = \frac{1}{6} + C, \text{so } C = -\frac{1}{6}$$
$$f(x) = \left(\frac{1}{6}\right)(e^{2x^3} - 1)$$

7 The velocity of an object is given by $v = t(t^2 - 3)^4$ m s^{-1}.
If the displacement is 0 after 2 s, find:

a the equation for the displacement

$$x(t) = \frac{(t^2 - 3)^5}{10} + C$$
$$0 = \frac{(1)^5}{10} + C$$
$$C = -\frac{1}{10}$$
$$\therefore x(t) = \left(\frac{1}{10}\right)[(t^2 - 3)^5 - 1]$$

b the displacement after 3 s.

$$x(3) = \left(\frac{1}{10}\right)[(6)^5 - 1] = 777.5 \text{ m}$$